Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 281 – Engineering Materials

Lecture 7: Phase Diagram

S. Boahen (PhD)

Lesson Objectives

After studying this chapter you should be able to do the following:

1. (a) Schematically sketch simple isomorphous and eutectic phase diagrams.

   (b) On these diagrams label the various phase regions.

   (c) Label liquidus, solidus, and solvus lines.

2. Given a binary phase diagram, the composition of an alloy, its temperature, and assuming that the alloy is at equilibrium, determine

(a) what phase(s) is (are) present,

(b) the composition(s) of the phase(s), and

(c) the mass fraction(s) of the phase(s).

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Learning Objectives

3. For some given binary phase diagram, do the following:

(a) locate the temperatures and compositions of all eutectic, eutectoid, peritectic, and congruent phase transformations; and

(b) write reactions for all these transformations for either heating or cooling.

4. Given the composition of an iron–carbon alloy containing between 0.022 wt% C and 2.14 wt% C, be able to

(a) specify whether the alloy is hypoeutectoid or hypereutectoid,

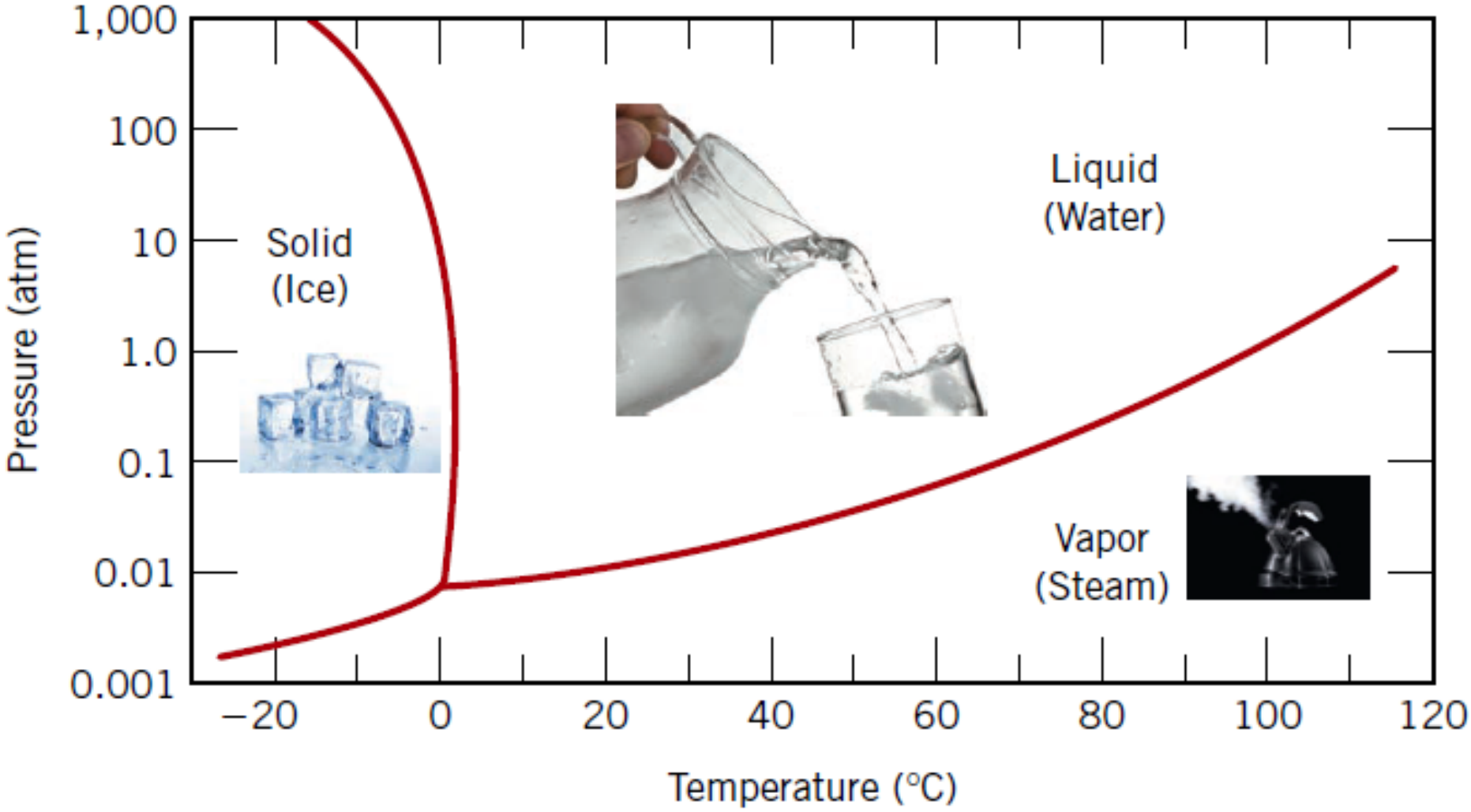
(b) name the proeutectoid phase,

(c) compute the mass fractions of proeutectoid phase and pearlite, and

(d) make a schematic diagram of the microstructure at a temperature just below the eutectoid.

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Phase Diagram of Water (H_20)



Introduction

- Phase diagrams provide valuable information about melting, casting, crystallization, and other phenomena. This chapter presents and discusses

✓ Terminology associated with phase diagrams and phase transformations

✓ Pressure–temperature phase diagrams for pure materials

✓ The interpretation of phase diagrams

✓ Some of the common and relatively simple binary phase diagrams, including that for the iron–carbon system

✓ The development of equilibrium microstructures, upon cooling, for several situations.

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Definitions and Basic Terminologies

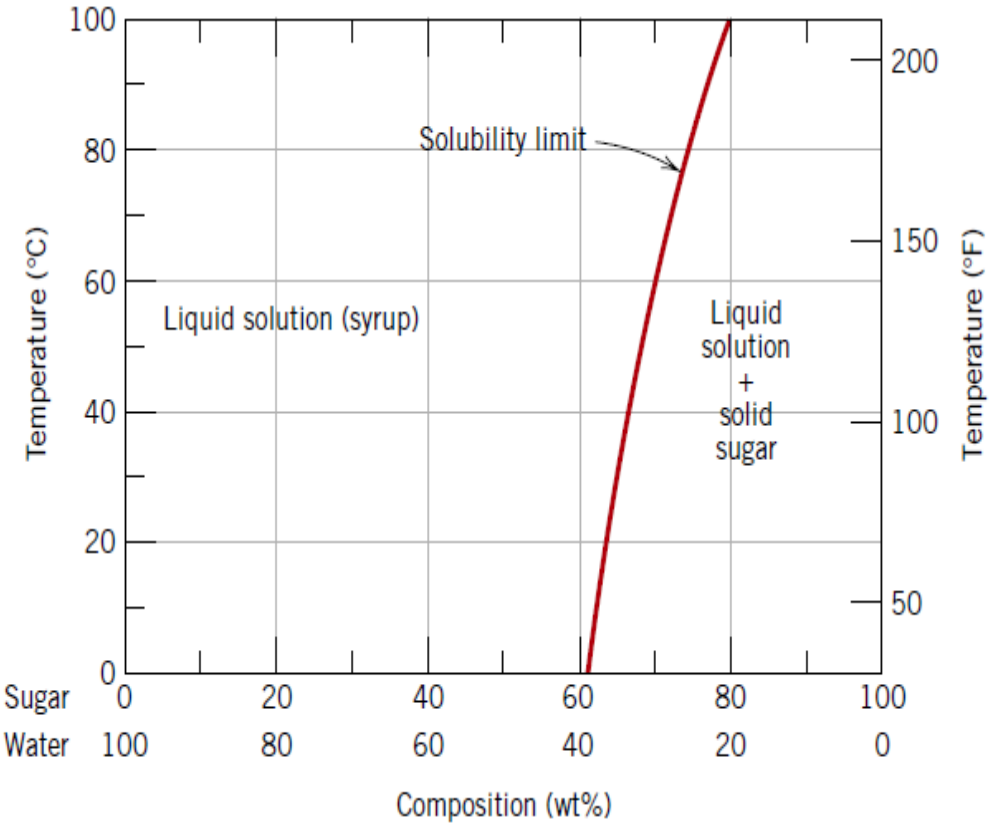
- **Components** are pure metals and/or compounds of which an alloy is composed. For example, in a copper–zinc brass, the components are Cu and Zn.
- **System** may refer to a specific body of material under consideration (e.g., a ladle of molten steel). Or it may relate to the series of possible alloys consisting of the same components, but without regard to alloy composition (e.g., the iron–carbon system).
- **Solid solution** consists of atoms of at least two different types; the solute atoms occupy either substitutional or interstitial positions in the solvent lattice, and the crystal structure of the solvent is maintained.

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Definitions and Basic Terminologies

- **Solubility limit:** For many alloy systems and at some specific temperature, there is a maximum concentration of solute atoms that may dissolve in the solvent to form a solid solution; this is called a **solubility limit.**

- The addition of solute in excess of this solubility limit results in the formation of another solid solution or compound that has a distinctly different composition.



Definitions and Basic Terminologies

- **Phases:** A phase may be defined as a homogeneous portion of a system that has uniform physical and chemical characteristics.
- Sometimes, a single-phase system is termed ***homogeneous***.
- Systems composed of two or more phases are termed ***mixtures*** or ***heterogeneous systems***.
- Most metallic alloys and, for that matter, ceramic, polymeric, and composite systems are heterogeneous.

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Definitions and Basic Terminologies

- **Microstructure:** The physical properties and, in particular, the mechanical behaviour of a material depend on the microstructure.
- Microstructure is subject to direct microscopic observation, using optical or electron microscopes.
- In metal alloys, microstructure is characterized by the number of phases present, their proportions, and the manner in which they are distributed or arranged.
- The microstructure of an alloy depends on such variables as the alloying elements present, their concentrations, and the heat treatment of the alloy (i.e., the temperature, the heating time at temperature, and the rate of cooling to room temperature).

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Definitions and Basic Terminologies

- **Phase equilibrium:** Refers to equilibrium as it applies to systems in which more than one phase may exist.
- Phase equilibrium is reflected by a constancy with time in the phase characteristics of a system.
- Mostly in solid systems, a state of equilibrium is never completely achieved because the rate of approach to equilibrium is extremely slow; such a system is said to be in a non-equilibrium or **metastable** state.
- A **metastable** state or microstructure may persist indefinitely, experiencing only extremely slight and almost imperceptible changes as time progresses.
- Often, metastable structures are of more practical significance than equilibrium ones.

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One - Component (OR UNARY) Phase Diagrams

- Information about the control of the phase structure of a particular system is conveniently and concisely displayed in a **phase diagram or** an ***equilibrium diagram***
- There are three externally controllable parameters that will affect phase structure: ***temperature, pressure, and composition***.
- Phase diagrams are constructed when various combinations of these parameters are plotted against one another.
- The simplest and easiest type of phase diagram to understand is that for a one-component system, in which composition is held constant (i.e., the phase diagram is for a pure substance).
- This means that pressure and temperature are the variables.

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One - Component (OR UNARY) Phase Diagrams

- One-component phase diagram (or *unary phase diagram*) [sometimes also called a *pressure–temperature* (or *P–T*) *diagram*] is represented as a two dimensional plot of pressure (ordinate, or vertical axis) versus temperature (abscissa, or horizontal axis).
- Most often, the pressure axis is scaled logarithmically.

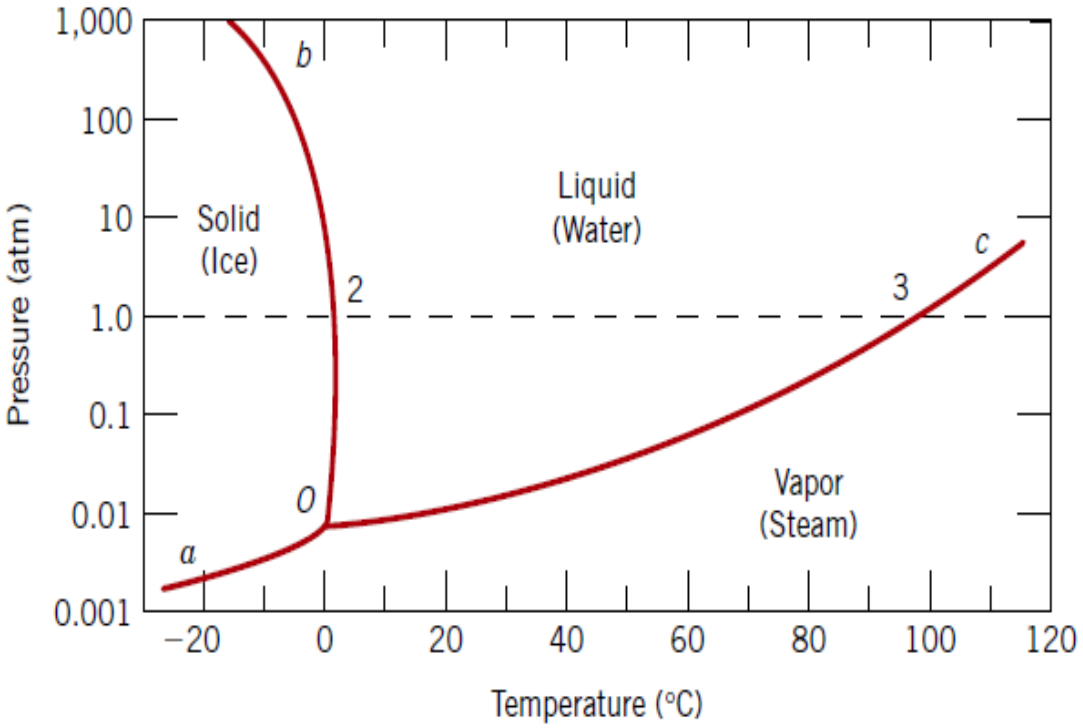


Fig. Pressure–temperature phase diagram for H2O.

Binary Phase

- Binary phase diagrams are maps that represent the relationships between temperature and the compositions and quantities of phases at equilibrium, which influence the microstructure of an alloy.
- Many microstructures develop from phase transformations, the changes that occur when the temperature is altered (ordinarily upon cooling). This may involve the transition from one phase to another, or the appearance or disappearance of a phase.
- Binary phase diagrams are helpful in predicting phase transformations and the resulting microstructures, which may have equilibrium or non-equilibrium character.

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Binary Phase Diagram

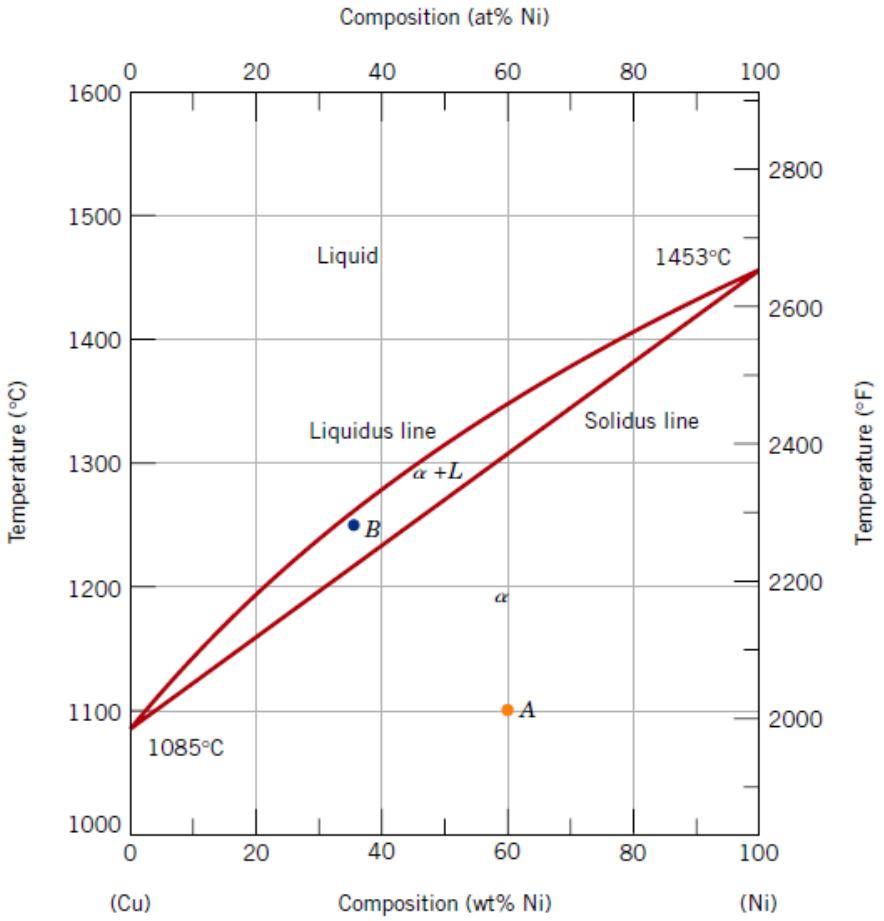


Fig. The copper–nickel phase diagram.

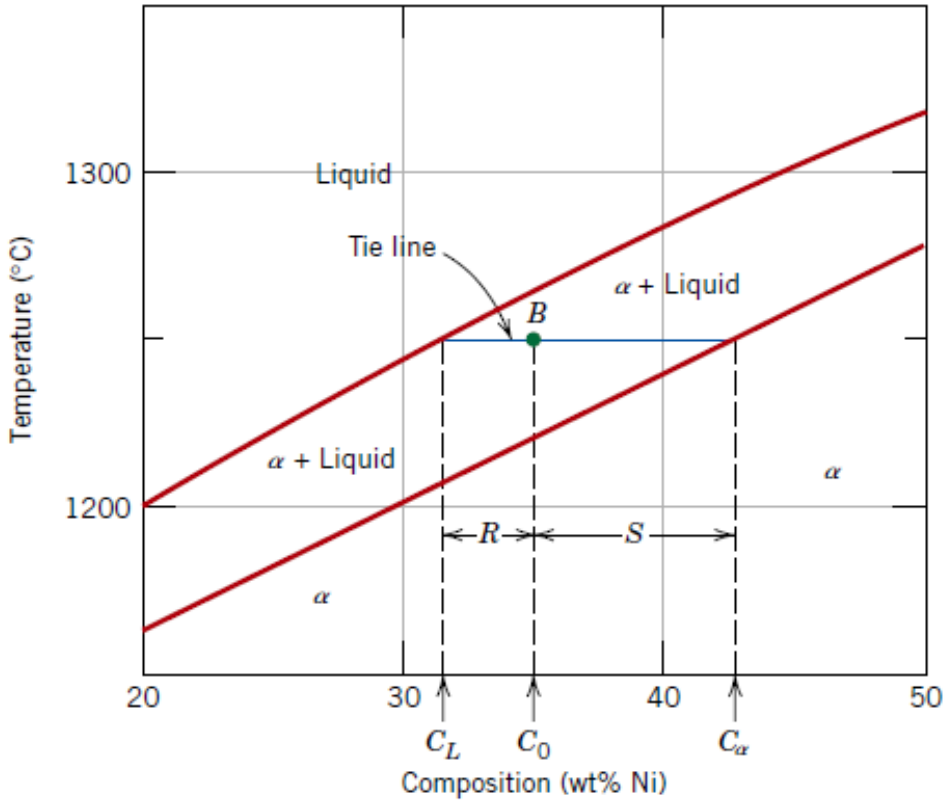


Fig. A portion of the copper–nickel phase diagram for which compositions and phase amounts are determined at point *B*

Interpretation of Phase Diagrams

- For a binary system of known composition and temperature that is at equilibrium, at least three kinds of information are available:

(1) the phases that are present

(2) the compositions of these phases

(3) the percentages or fractions of the phases.

Interpretation of Phase Diagrams: Determination of Phases Present

- Locate the temperature - composition point on the diagram and note the phase(s) with which the corresponding phase field is labelled.
- For example, an alloy of composition 60 wt% Ni–40 wt% Cu at 1100°C would be located at point *A as shown* in Fig.
- Because this is within the α region, only the single α phase will be present.
- On the other hand, a 35 wt% Ni–65 wt% Cu alloy at 1250C (point *B*) will consist of both α and liquid phases at equilibrium.

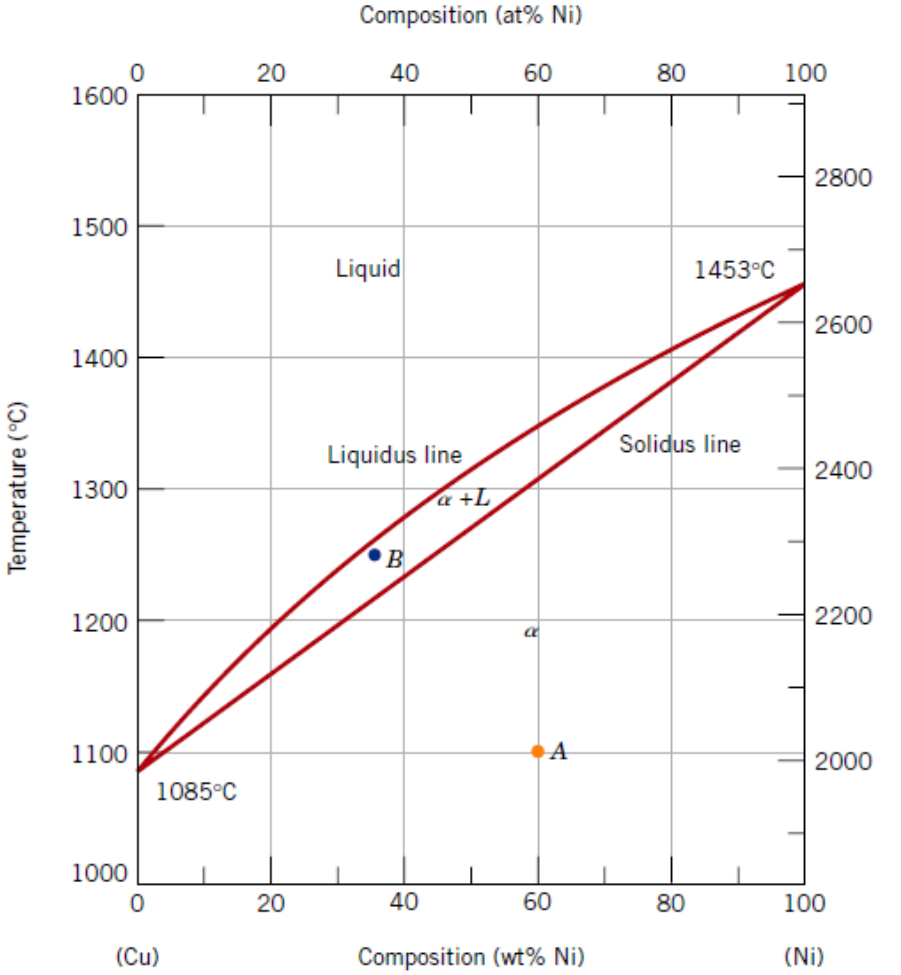


Fig. The copper–nickel phase diagram.

Interpretation of Phase Diagrams: Determination of Phase Compositions

- Different methods are used for single- and two-phase regions.
- If only **one phase** is present, the composition of this phase is simply the same as the overall composition of the alloy.
- For example, consider the 60 wt% Ni–40 wt% Cu alloy at 1100C (point *A*). At this composition and temperature, only the α phase is present, having a composition of 60 wt% Ni–40 wt% Cu.

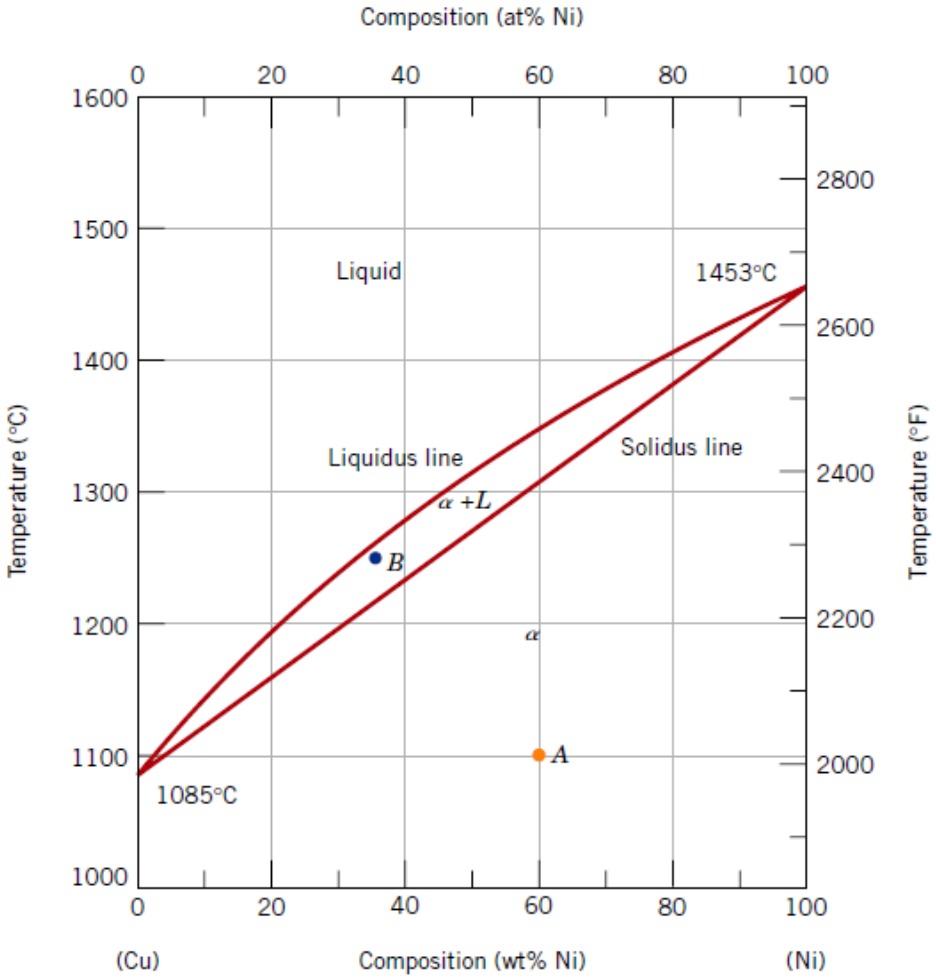


Fig. 3 The copper–nickel phase diagram.

Interpretation of Phase Diagrams: Determination of 2 Phase Compositions

- To compute the equilibrium concentrations of the two phases, the following procedure is used:

a. A tie line is constructed across the two-phase region at the temperature of the alloy.

b. The intersections of the tie line and the phase boundaries on either side are noted.

c. Perpendiculars are dropped from these intersections to the horizontal composition axis, from which the composition of each of the respective phases is read.

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Interpretation of Phase Diagrams: Determination of 2 Phase Compositions

- For example, consider again the 35 wt% Ni–65 wt% Cu alloy at 1250℃, located at point *B* in Fig.4 and lying within the $\alpha + L$ region. Thus, the problem is to determine the composition (in wt% Ni and Cu) for both the α and liquid phases. The tie line has been constructed across the $\alpha + L$ phase region, as shown in Figure.
- The perpendicular from the intersection of the tie line with the liquidus boundary meets the composition axis at 31.5 wt% Ni–68.5 wt% Cu, which is the composition of the liquid phase, C_L.
- Likewise, for the solidus–tie line intersection, we find a composition for the α solid-solution phase, C_a, of 42.5 wt% Ni–57.5 wt% Cu.

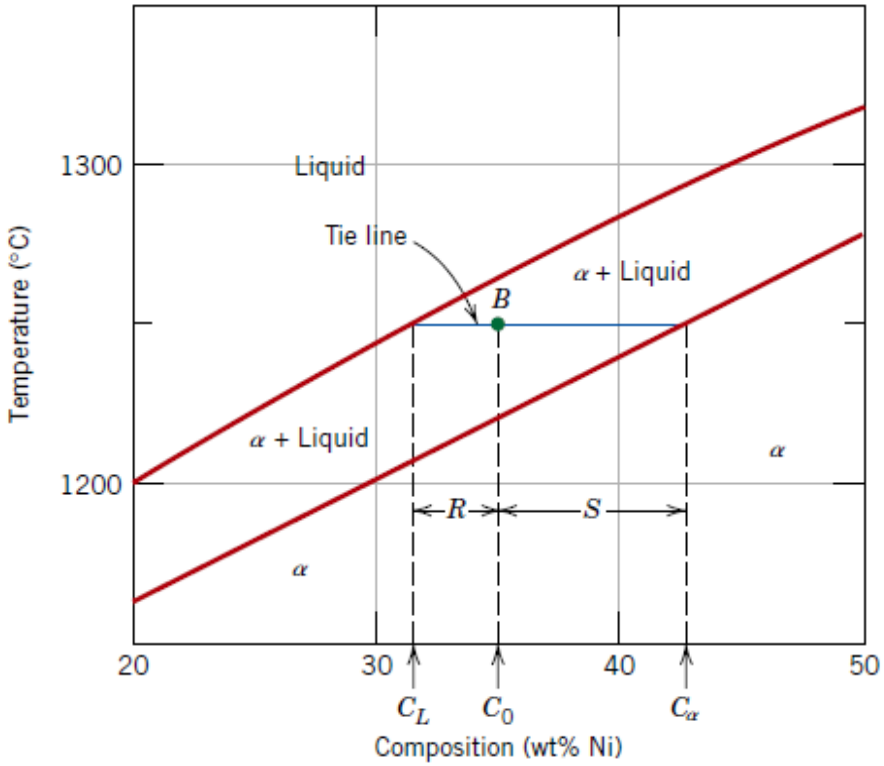


Fig. 4 A portion of the copper–nickel phase diagram for which compositions and phase amounts are determined at point *B*

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Interpretation of Phase Diagrams: Determination of Phase Amounts

- The relative amounts (as fraction or as percentage) of the phases present at equilibrium can be computed with the aid of phase diagrams.
- In a two-phase region, the tie line is used in conjunction with the **lever rule** as follows:

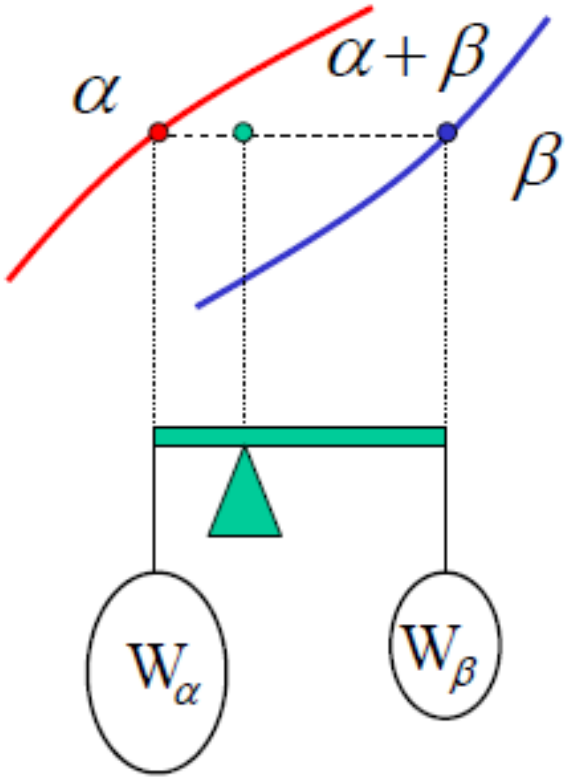
1. The tie line is constructed across the two-phase region at the temperature of the alloy.
2. The overall alloy composition is located on the tie line.
3. The fraction of one phase is computed by taking the length of tie line from the overall alloy composition to the phase boundary for the *other* phase, and dividing by the total tie line length.
4. The fraction of the other phase is determined in the same manner.
5. If phase percentages are desired, each phase fraction is multiplied by 100. When the composition axis is scaled in weight percent, the phase fractions computed using the lever rule are mass fractions—the mass (or weight) of a specific phase divided by the total alloy mass (or weight). The mass of each phase is computed from the product of each phase fraction and the total alloy mass.

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The lever rule: Finding the amounts of phases in a two phase region

The lever rule is a mechanical analogy to the mass balance calculation. The tie line in the two-phase region is analogous to a lever balanced on a fulcrum.

1. Locate composition and temperature in diagram
2. In two phase region draw the tie line or isotherm
3. Fraction of a phase is determined by taking the length of the tie line to the phase boundary for the other phase, and dividing by the total length of tie line



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Derivation of the Lever Rule

1) All material must be in one phase or the other:

$$W_\alpha + W_L = 1$$

2) Mass of a component that is present in both phases equal to the mass of the component in one phase + mass of the component in the second phase:

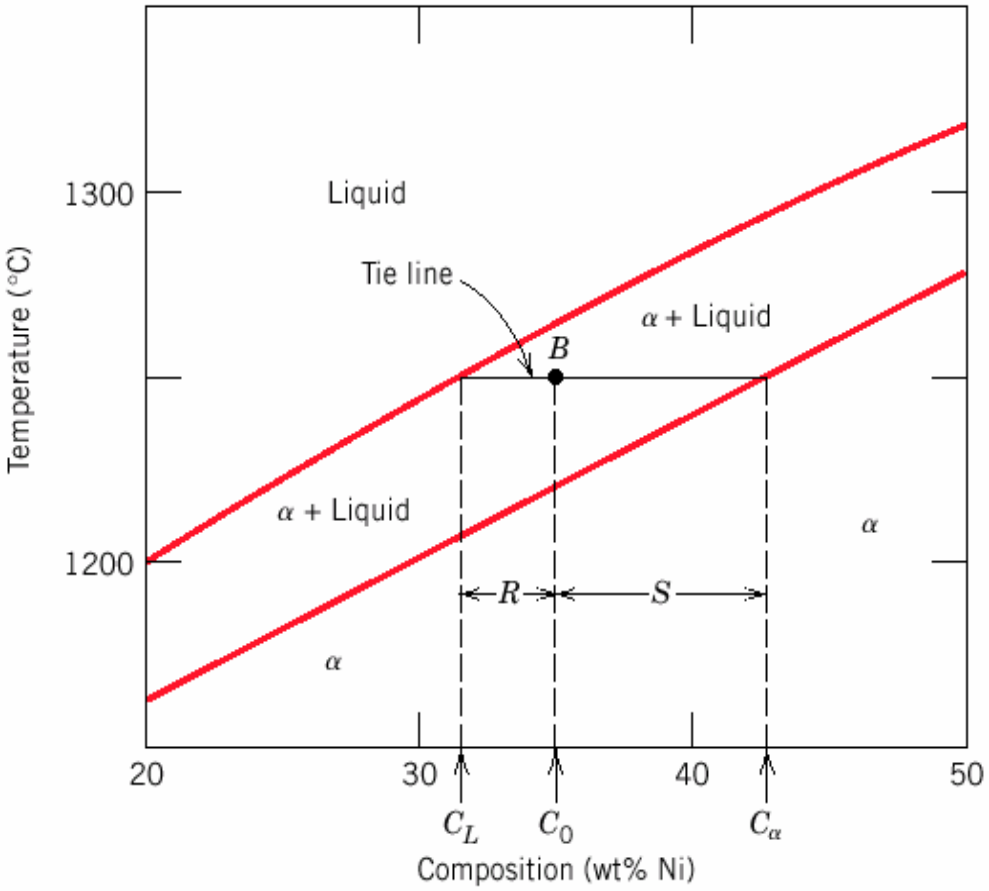
$$W_\alpha C_\alpha + W_L C_L = C_o$$

3) Solution of these equations gives us the lever rule.

$$W_L = (C_\alpha - C_o)/(C_\alpha - C_o)$$

$$W_\alpha = (C_o - C_L)/(C_\alpha - C_L)$$

Example: Phase compositions and amounts



$C_o = 35$ wt. %, $C_L = 31.5$ wt. %, $C_\alpha = 42.5$ wt. %

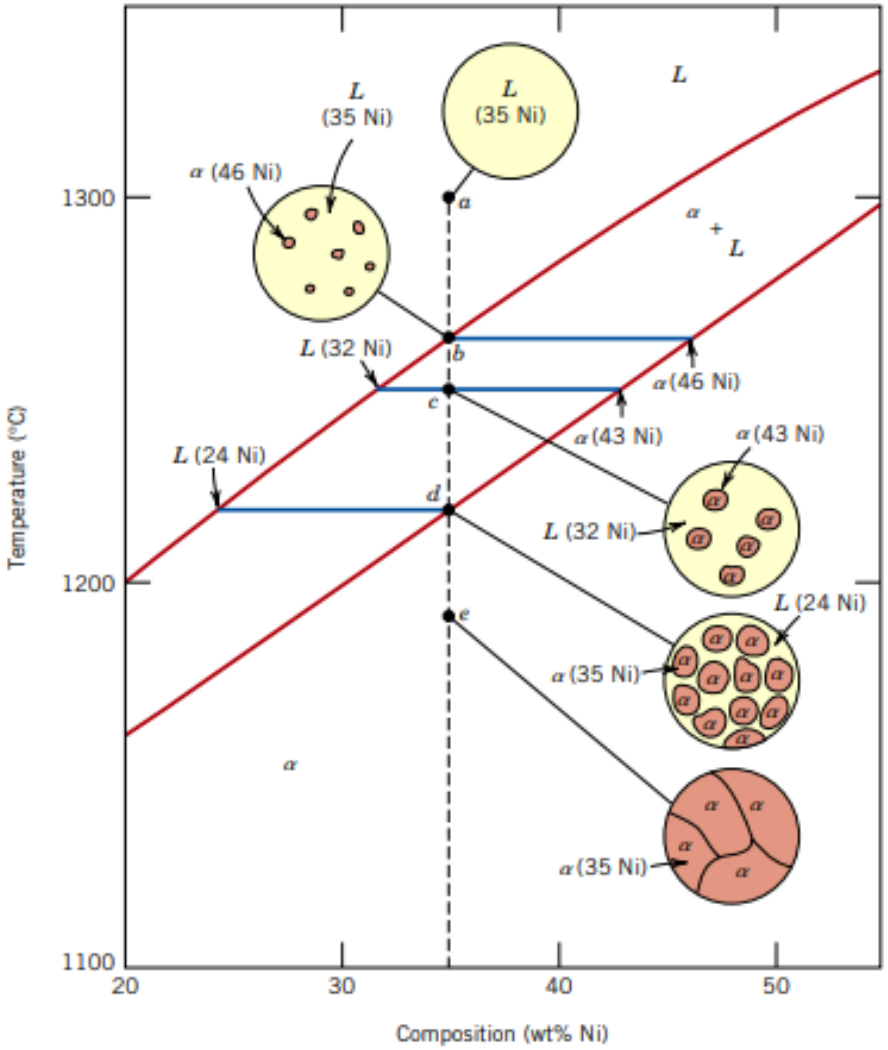
Mass fractions:

$$W_L = S / (R+S) = (C_\alpha - C_o) / (C_\alpha - C_L) = 0.68$$

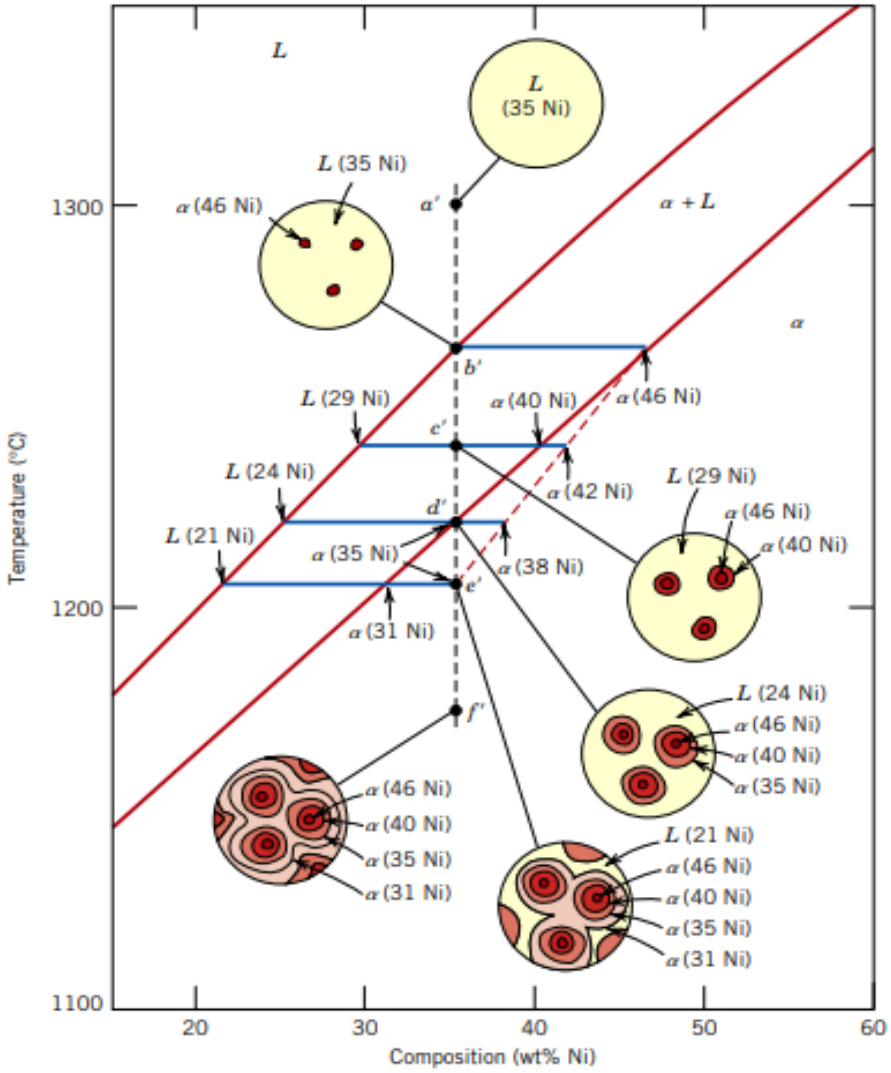
$$W_\alpha = R / (R+S) = (C_o - C_L) / (C_\alpha - C_L) = 0.32$$

Development of microstructure in isomorphous alloys: Equilibrium (Very slow) Cooling

- Solidification in the solid + liquid phase occurs gradually upon cooling from the liquidus line.
- The composition of the solid and the liquid change gradually during cooling (as can be determined by the tie-line method.)
- Nuclei of the solid phase form and they grow to consume all the liquid at the solidus line.



Development of microstructure in isomorphous alloys: Non-equilibrium (Fast) cooling



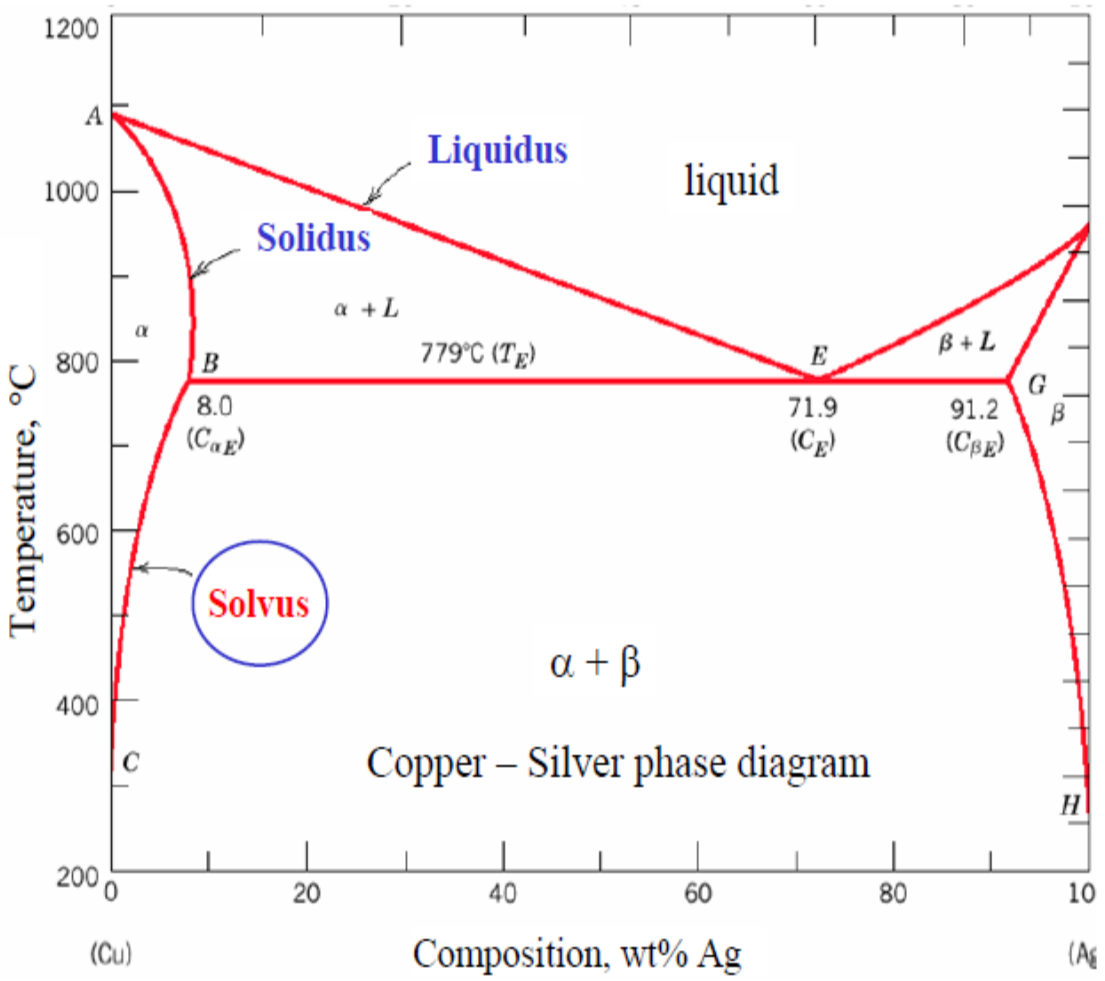
Development of microstructure in isomorphous alloys: Non-equilibrium (Fast) cooling

- Compositional changes require diffusion in solid and liquid phases
- Diffusion in the solid state is very slow. ⇒ The new layers that solidify on top of the existing grains have the equilibrium composition at that temperature but once they are solid their composition does not change. ⇒ Formation of layered (cored) grains and the invalidity of the tie-line method to determine the composition of the solid phase.
- The tie-line method still works for the liquid phase, where diffusion is fast. Average Ni content of solid grains is higher. ⇒ Application of the lever rule gives us a greater proportion of liquid phase as compared to the one for equilibrium cooling at the same T. ⇒ Solidus line is shifted to the right (higher Ni contents), solidification is complete at lower T, the outer part of the grains are richer in the low-melting component (Cu).
- Upon heating grain boundaries will melt first. This can lead to premature mechanical failure.

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Binary Eutectic Systems

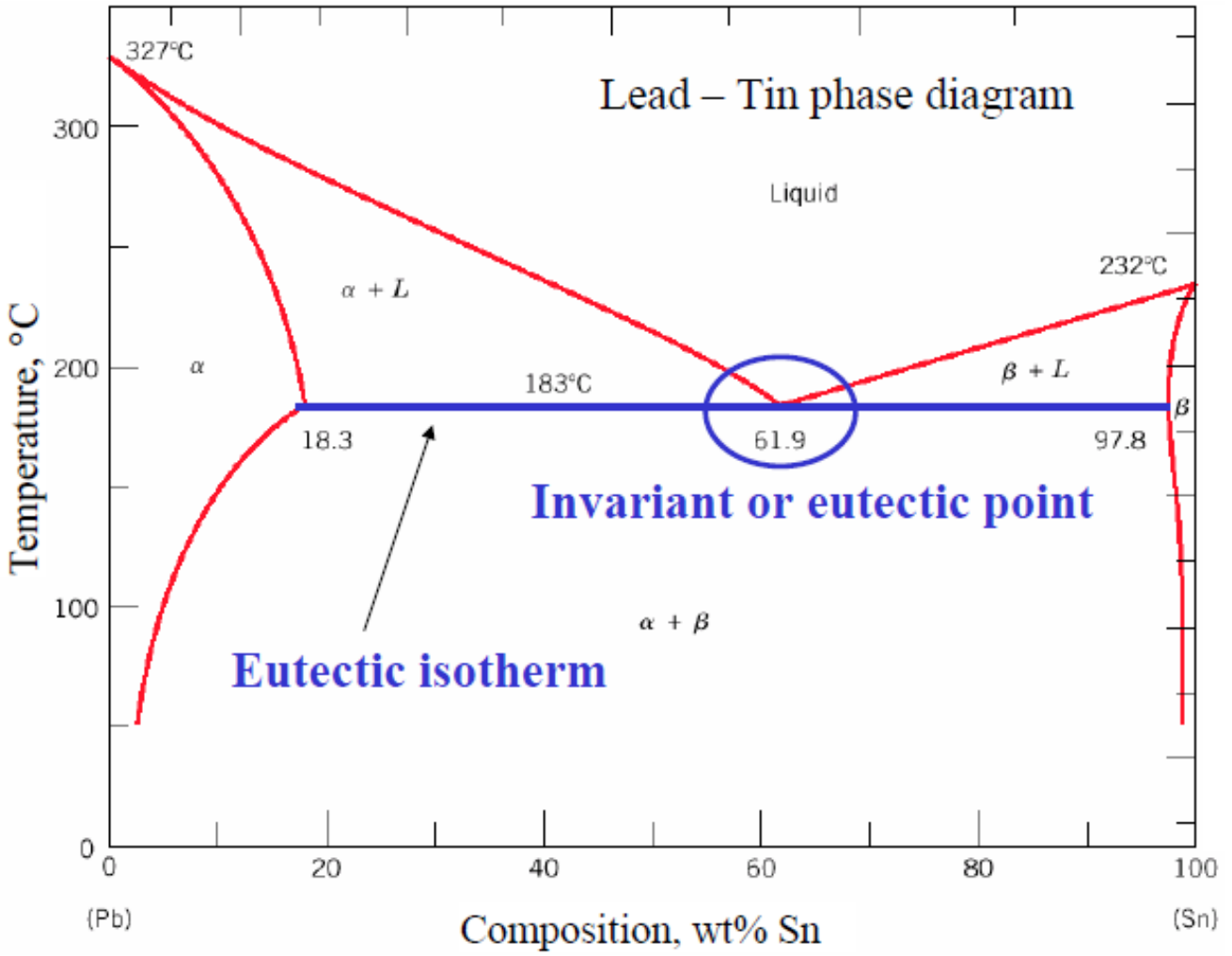
- Three single phase regions (α - solid solution of Ag in Cu matrix, β = solid solution of Cu in Ag matrix, L - liquid)
- Three two-phase regions (α + L, β +L, α +β)
- **Solvus** line separates one solid solution from a mixture of solid solutions. **Solvus line shows limit of solubility**



Binary Eutectic Systems

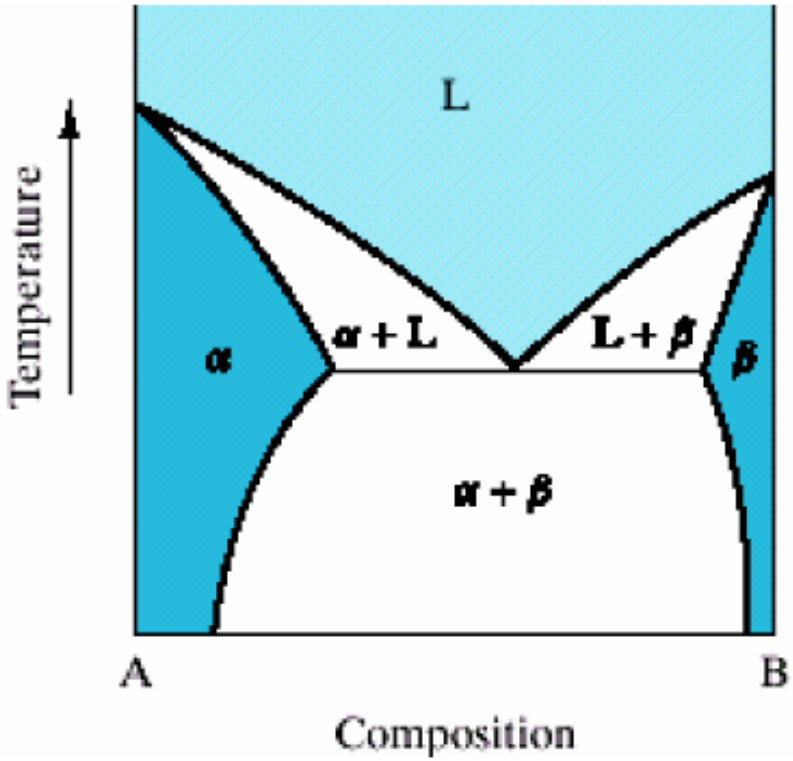
- **Eutectic or invariant point** - Liquid and two solid phases co-exist in equilibrium at the eutectic composition C_E and the eutectic temperature T_E.

- **Eutectic isotherm** - the horizontal solidus line at T_E.



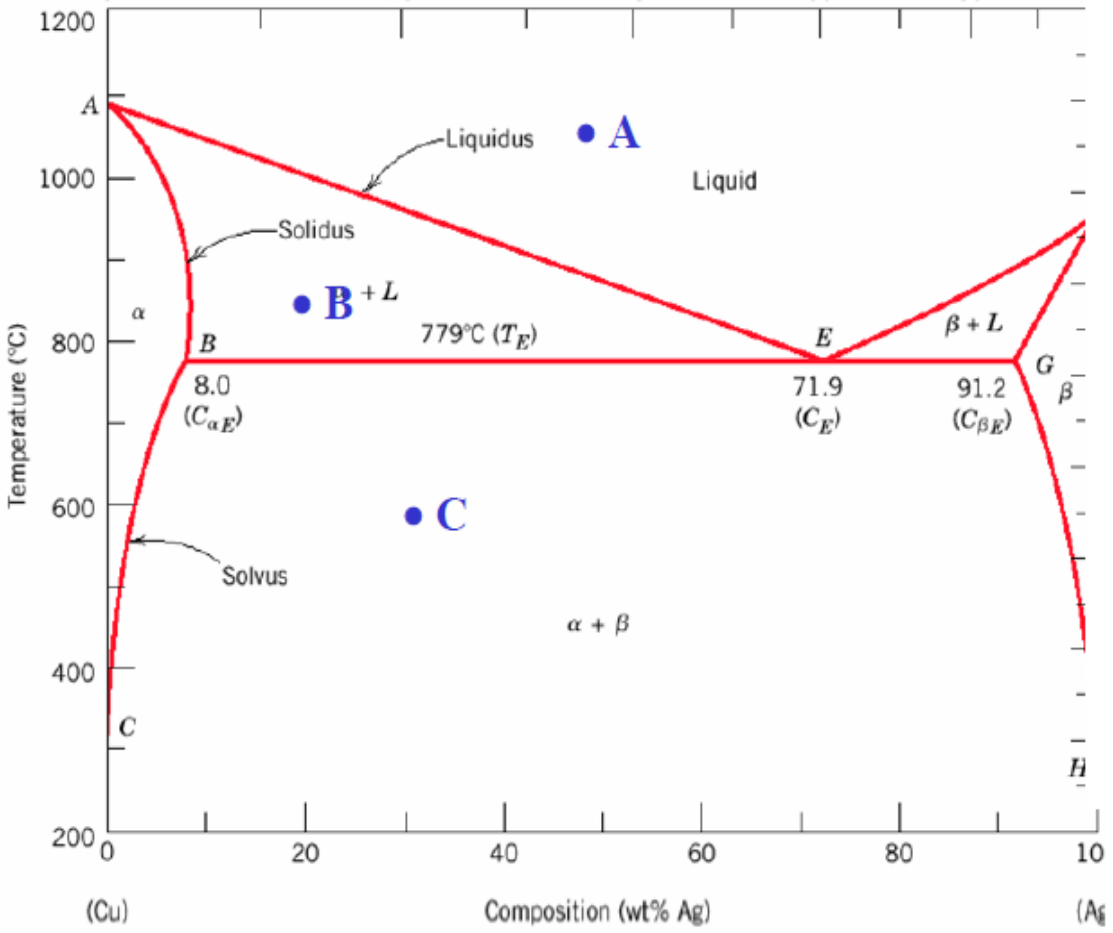
Binary Eutectic Systems

- **Eutectic reaction** – transition between liquid and mixture of two solid phases, α + β at eutectic concentration C_E.
- The melting point of the eutectic alloy is lower than that of the components (***eutectic = easy to melt in Greek***).
- At most two phases can be in equilibrium within a phase field.
- Three phases (L, α, β) may be in equilibrium only at a few points along the eutectic isotherm.
- Single-phase regions are separated by 2-phase regions.



Binary Eutectic Systems

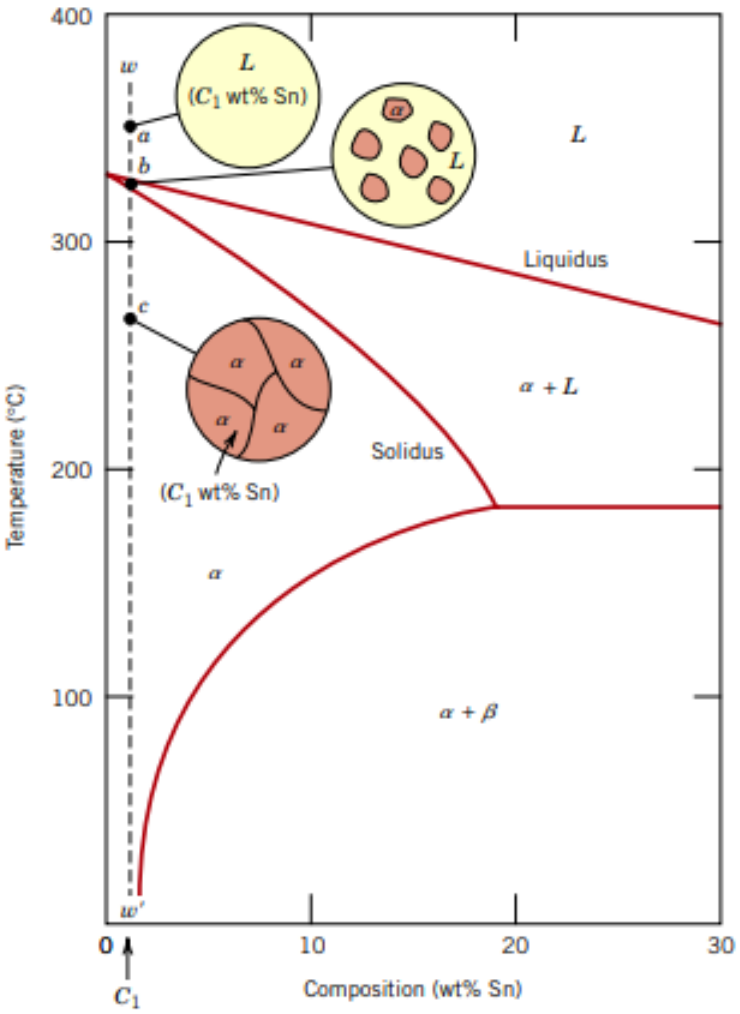
- Compositions and relative amounts of phases are determined from the same tie lines and lever rule, as for isomorphous alloys.
- For points A, B, and C calculate the compositions (wt. %) and relative amounts (mass fractions) of phases present.



Development of microstructure in eutectic alloys

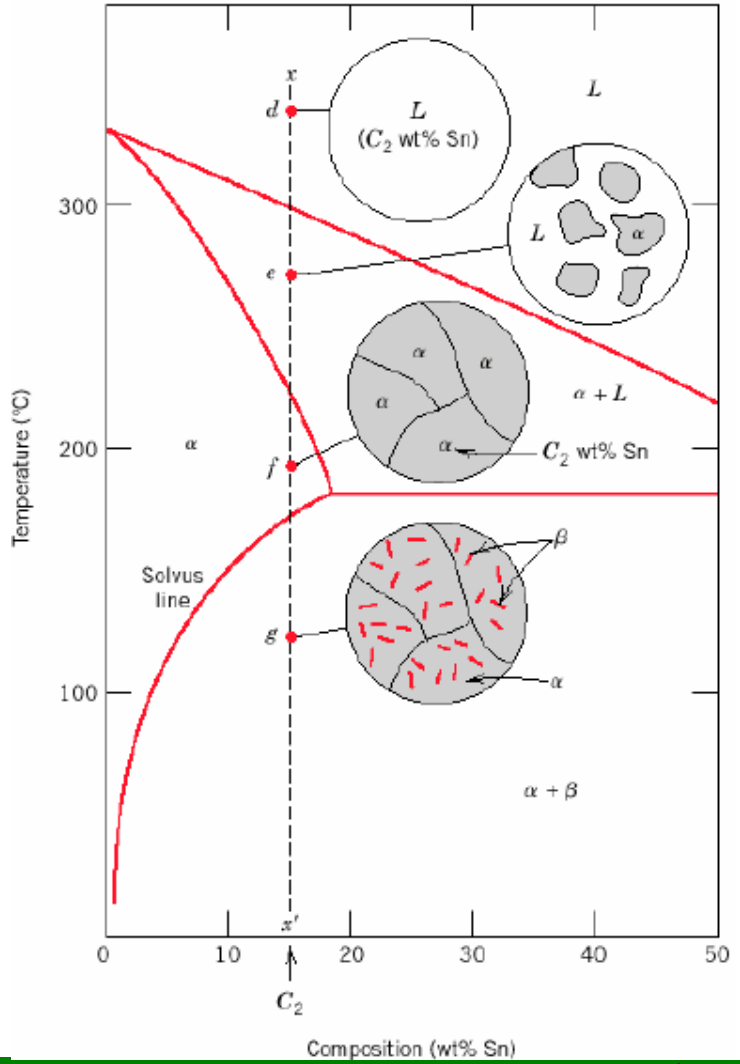
- Several different types of microstructure can be formed in slow cooling at different compositions.
- Let's consider cooling of liquid lead – tin system at different compositions.
- In this case of lead-rich alloy (0-2 wt. % of tin) solidification proceeds in the same manner as for isomorphous alloys (e.g. Cu-Ni) that we discussed earlier.

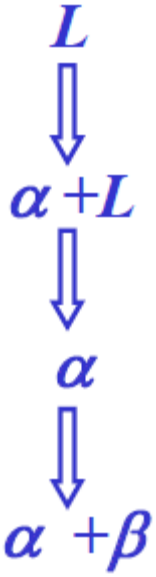
$$L \rightarrow \alpha + L \rightarrow \alpha$$



Development of microstructure in eutectic alloys

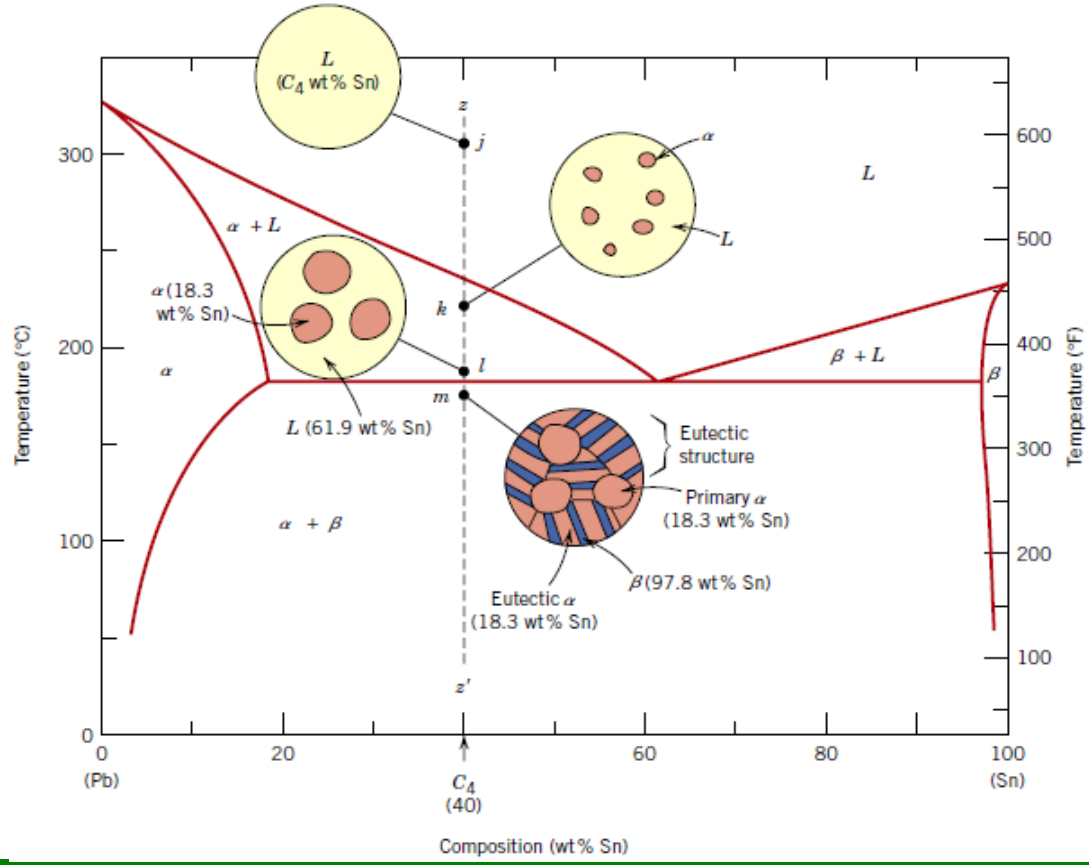
- At compositions between the room temperature solubility limit and the maximum solid solubility at the eutectic temperature, β phase nucleates as the α solid solubility is exceeded upon crossing the solvus line.





Development of microstructure in eutectic alloys

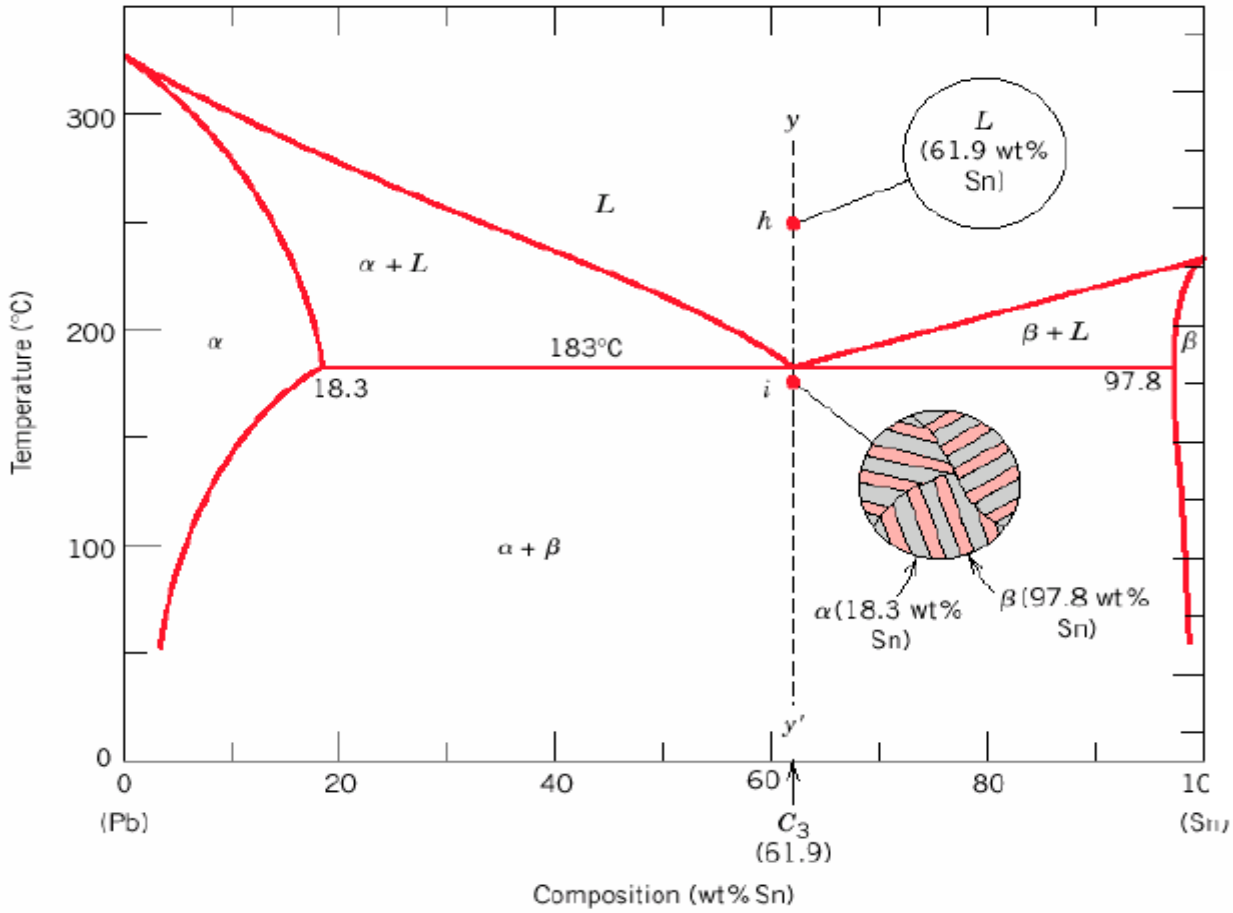
- **Primary** α phase is formed in the α + L region, and the eutectic structure that includes layers of α and β phases (called **eutectic** α and **eutectic** β phases) is formed upon crossing the eutectic isotherm.



$$L \rightarrow \alpha + L \rightarrow \alpha + \beta$$

Development of microstructure in eutectic alloys: Solidification at the eutectic composition

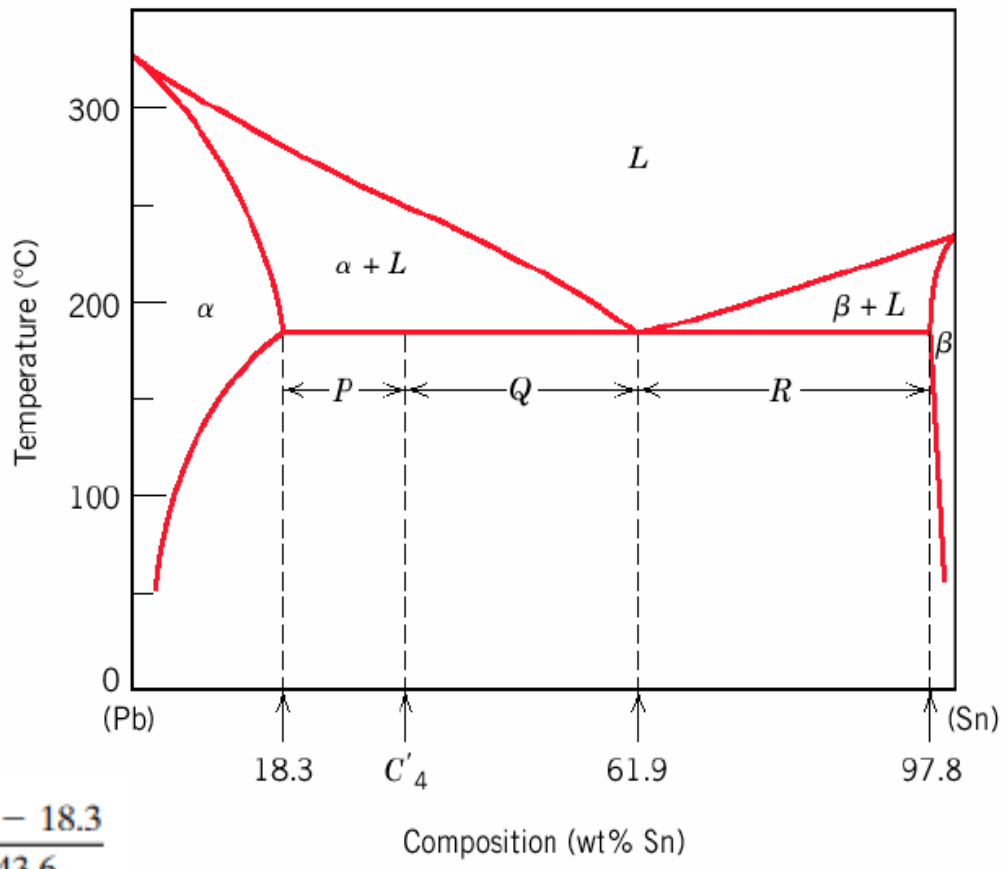
- No changes above the eutectic temperature T_E. At T_E the liquid transforms to α and β phases (eutectic reaction).
-



$L \rightarrow \alpha + \beta$

How to calculate relative amounts of micro-constituents?

- Eutectic micro-constituent forms from liquid having eutectic composition (61.9 wt% Sn)
- We can treat the eutectic as a separate phase and apply the lever rule to find the relative fractions of primary α phase (18.3 wt% Sn) and the eutectic structure (61.9 wt% Sn):



$$W_e = P / (P+Q) \text{ (eutectic)} = \frac{C_4' - 18.3}{61.9 - 18.3} = \frac{C_4' - 18.3}{43.6}$$

$$W_{\alpha'} = Q / (P+Q) \text{ (primary)} = \frac{61.9 - C_4'}{61.9 - 18.3} = \frac{61.9 - C_4'}{43.6}$$

How to calculate the total amounts of phases?

Fraction of α phase determined by application of the lever rule across the entire α + β phase field:

(α phase)

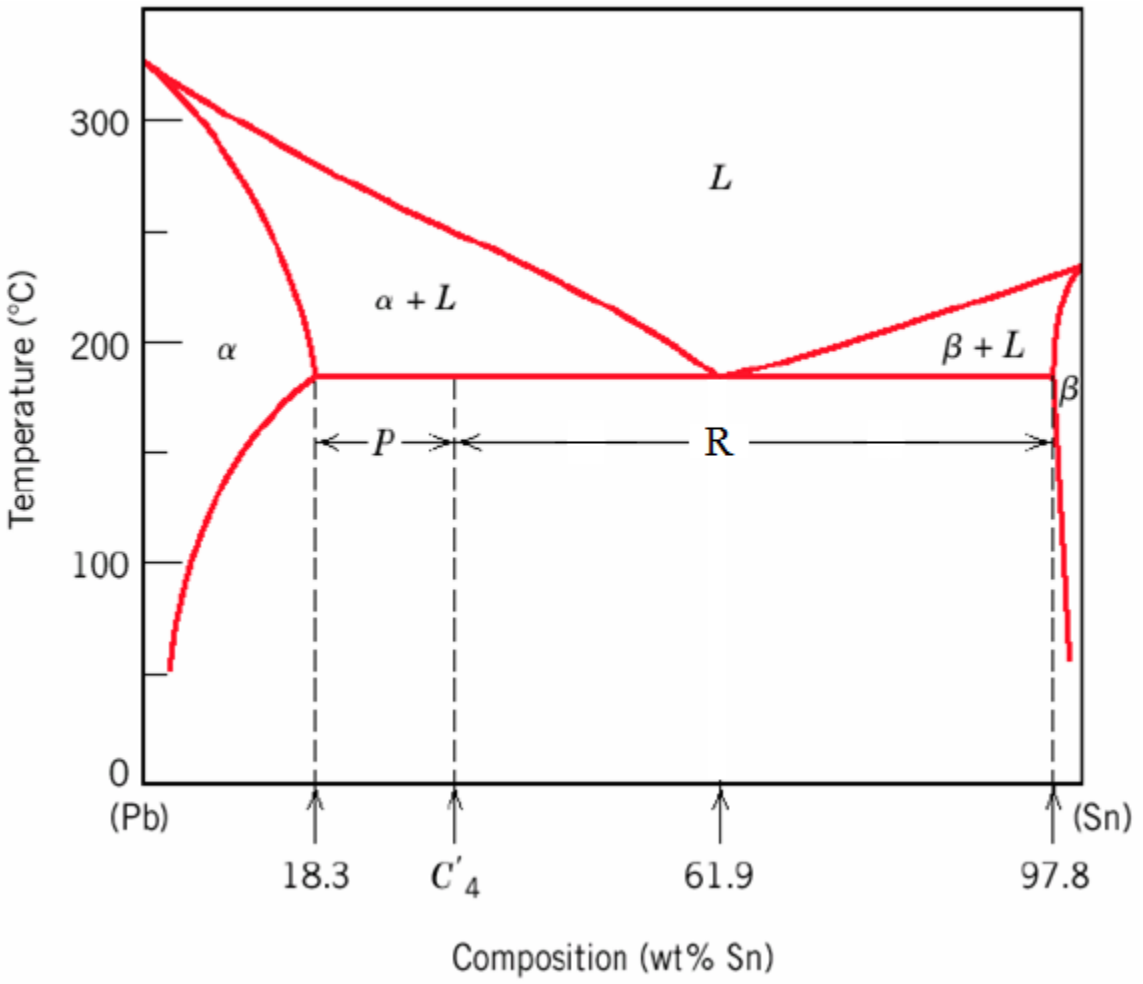
W_α = R / (P+R)

$$= \frac{97.8 - C'_4}{97.8 - 18.3} = \frac{97.8 - C'_4}{79.5}$$

(β phase)

W_β = P / (P+R)

$$= \frac{C'_4 - 18.3}{97.8 - 18.3} = \frac{C'_4 - 18.3}{79.5}$$



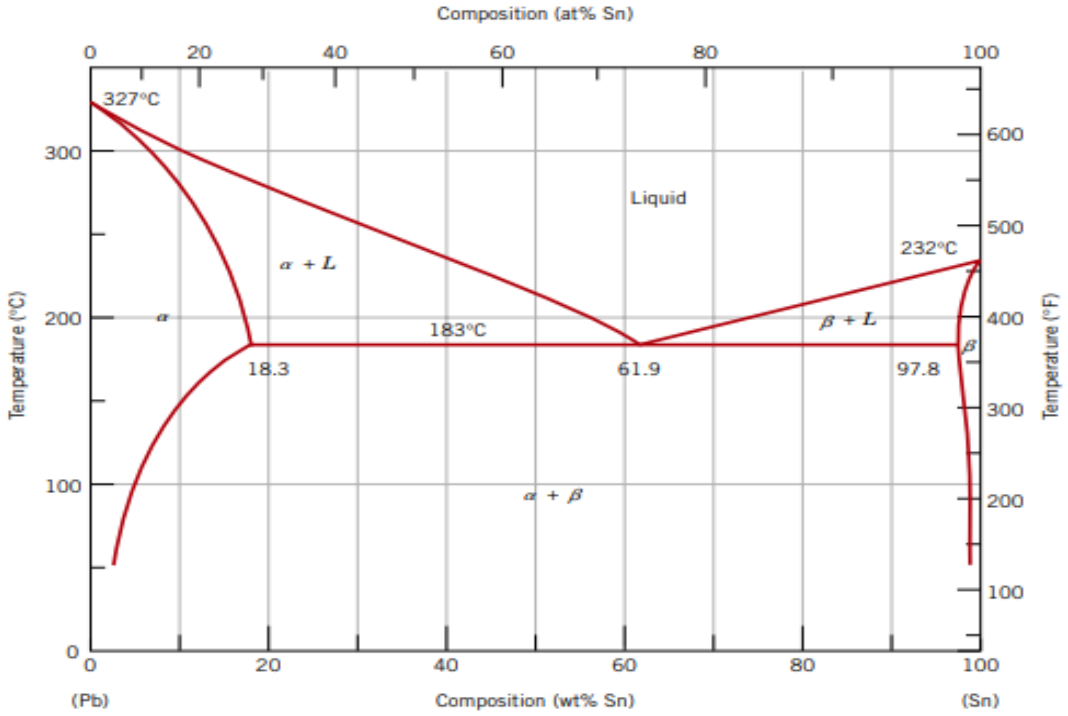
Example

For a 40 wt% Sn–60 wt% Pb alloy at 150°C (300°F) as shown in Fig. below,

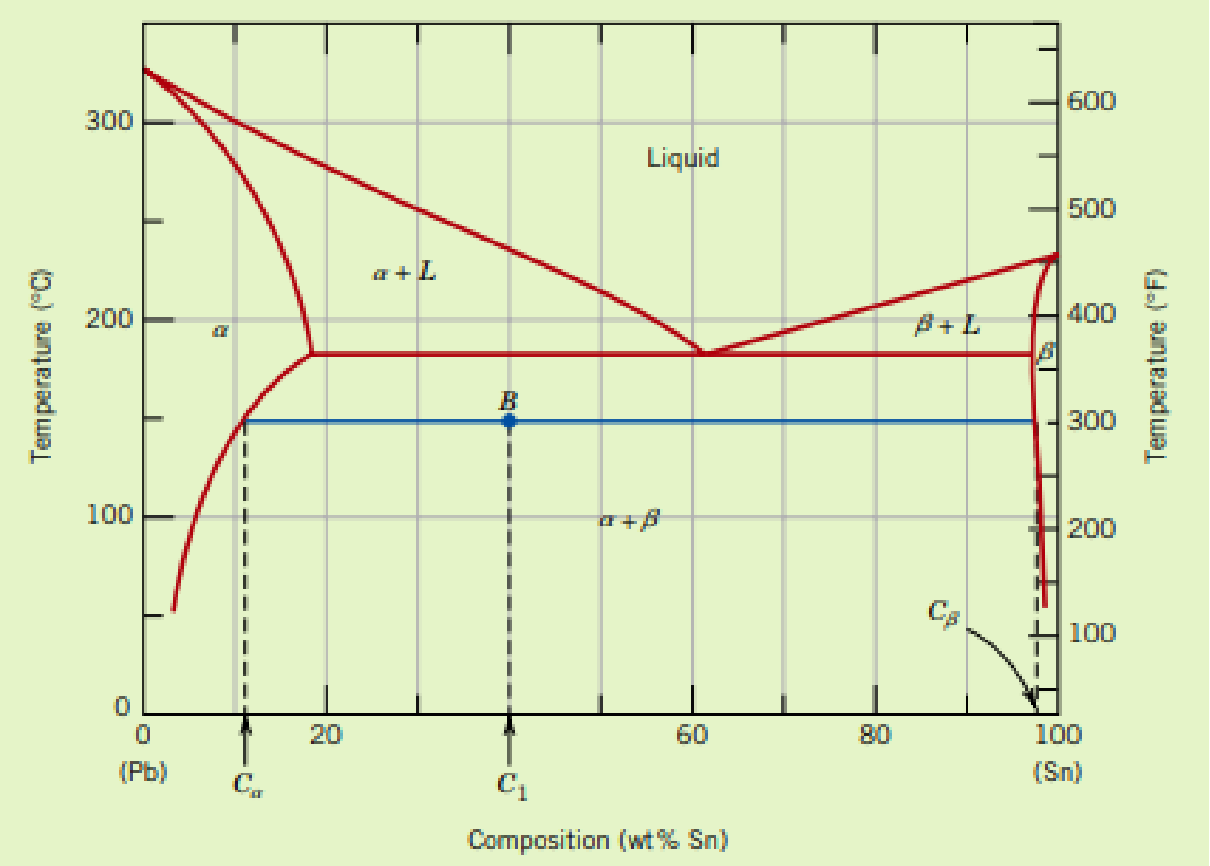
(a) what phase(s) is (are) present?

(b) what is (are) the composition(s) of the phase(s)?

(c) Calculate the relative amount of each phase present in terms of mass fraction



Solution



(a) Locate this temperature–composition point on the phase diagram (point B). Inasmuch as it is within the $\alpha + \beta$ region, both α and β phases will coexist.

Solution Cont.

(b) Because two phases are present, it becomes necessary to construct a tie line across the $\alpha + \beta$ phase field at 150℃, as indicated in Figure. The composition of the α phase corresponds to the tie line intersection with the $\alpha/(\alpha + \beta)$ solvus phase boundary - about 11 wt% Sn–89 wt% Pb, denoted as C_α. Similarly for the β phase, which will have a composition of approximately 98 wt% Sn–2 wt% Pb (C_β).

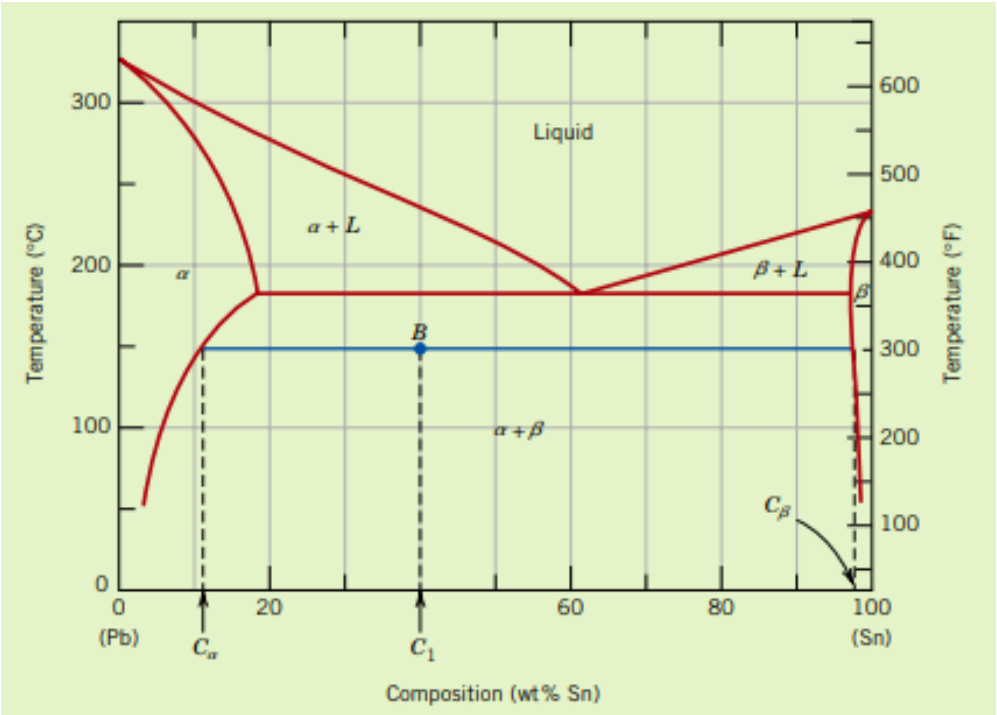
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Solution cont.

c) Because the alloy consists of two phases, it is necessary to employ the lever rule. If C_1 denotes the overall alloy composition, mass fractions may be computed by subtracting compositions, in terms of weight percent tin, as follows:

$$W_\alpha = \frac{C_\beta - C_1}{C_\beta - C_\alpha} = \frac{98 - 40}{98 - 11} = 0.67$$

$$W_\beta = \frac{C_1 - C_\alpha}{C_\beta - C_\alpha} = \frac{40 - 11}{98 - 11} = 0.33$$

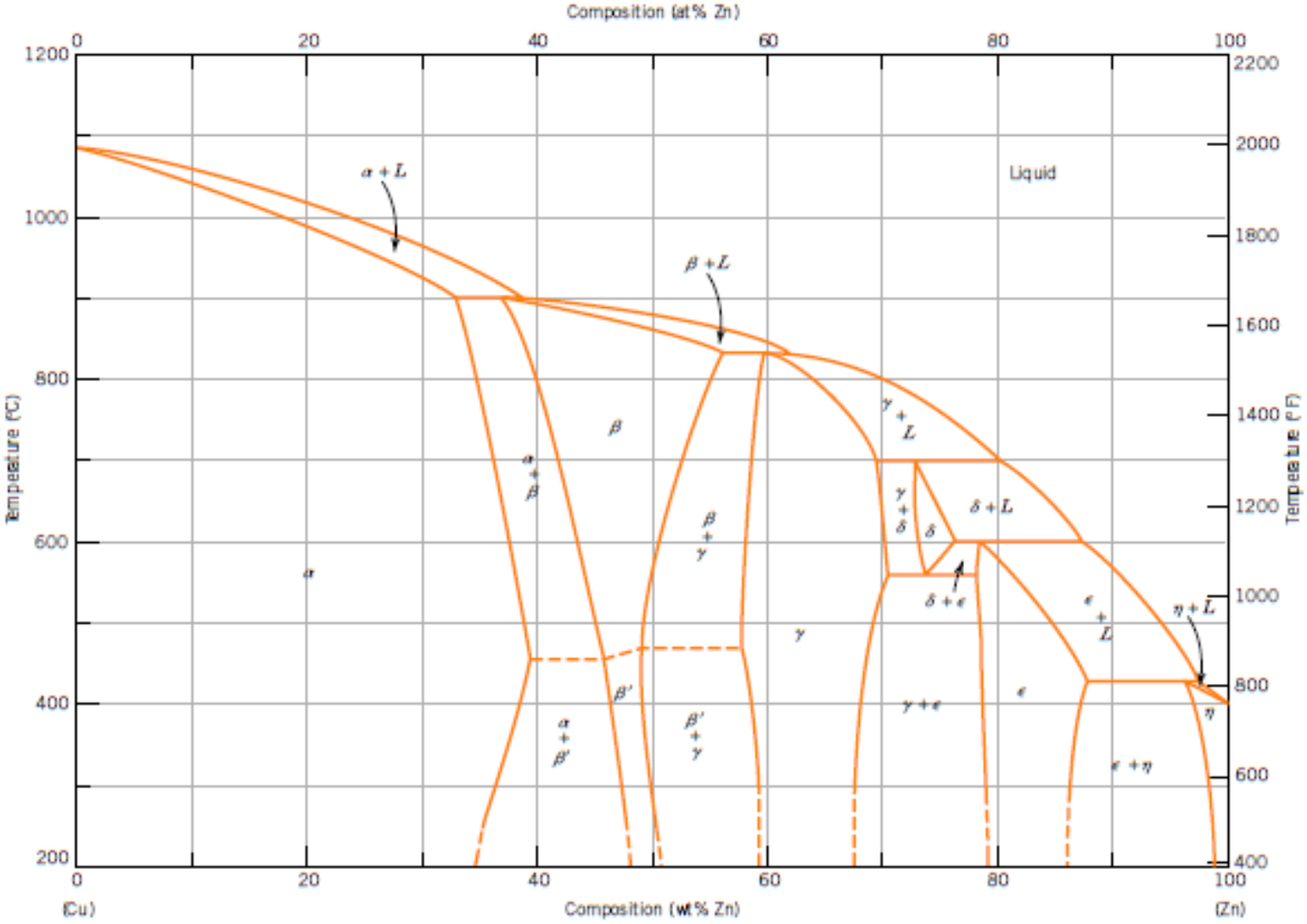


Phase diagrams with intermediate phases

- Eutectic systems that we have studied so far have only two solid phases (α and β) that exist near the ends of phase diagrams. These phases are called **terminal solid solutions.**
- Some binary alloy systems have **intermediate solid solution phases**. In phase diagrams, these phases are separated from the composition extremes (0% and 100%).
- Example: in Cu-Zn, α and η are terminal solid solutions, β, β', γ, δ, ε are intermediate solid solutions.

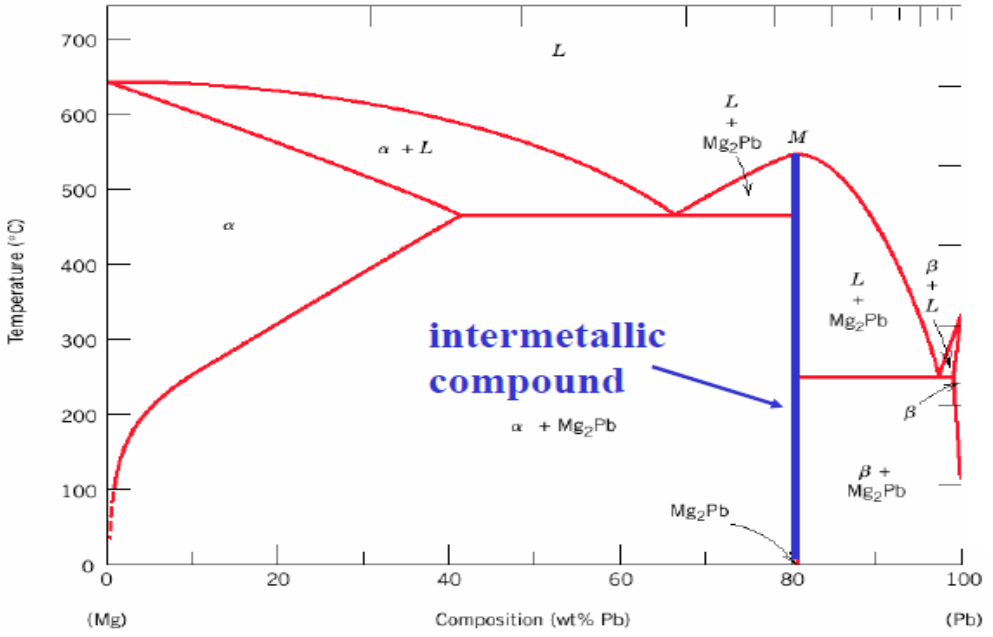
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Phase diagrams with intermediate phases



Phase diagrams with intermetallic compounds

- Besides solid solutions, **intermetallic compounds**, that have precise chemical compositions can exist in some systems.
- When using the lever rules, intermetallic compounds are treated like any other phase, except they appear not as a wide region but as a vertical line.
- This diagram can be thought of as two joined eutectic diagrams, for Mg-Mg_2Pb and Mg_2Pb-Pb. In this case compound Mg_2Pb can be considered as a component.

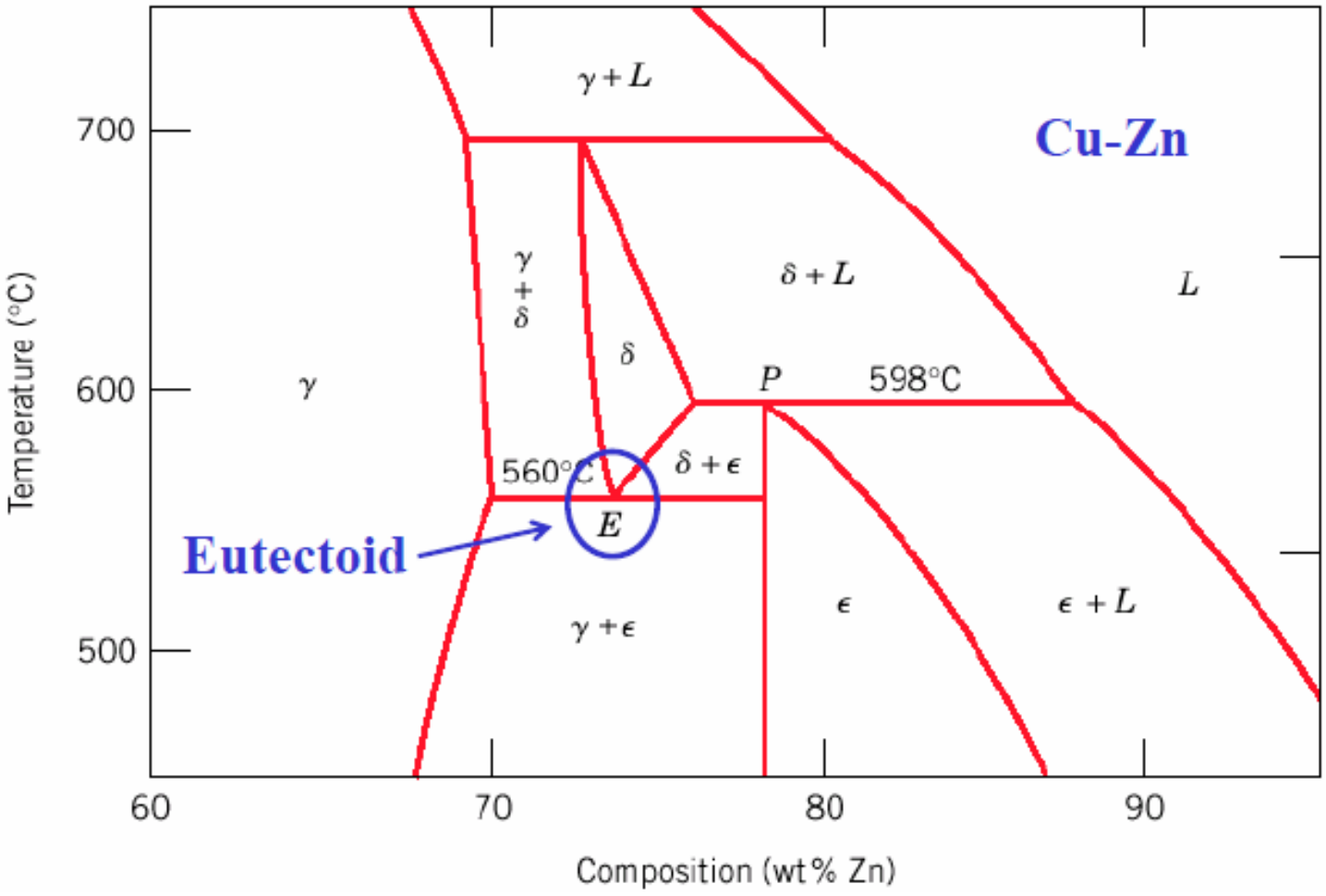


Eutectoid Reactions

- The **eutectoid** (*eutectic-like* in Greek) reaction is similar to the eutectic reaction but occurs from one solid phase to two new solid phases.
- Invariant point (the eutectoid) – three **solid** phases are in equilibrium.
- Upon cooling, a solid phase transforms into two other solid phases ($\delta \leftrightarrow \gamma + \varepsilon$)
- Looks as V on top of a horizontal tie line (eutectoid isotherm) in the phase diagram.

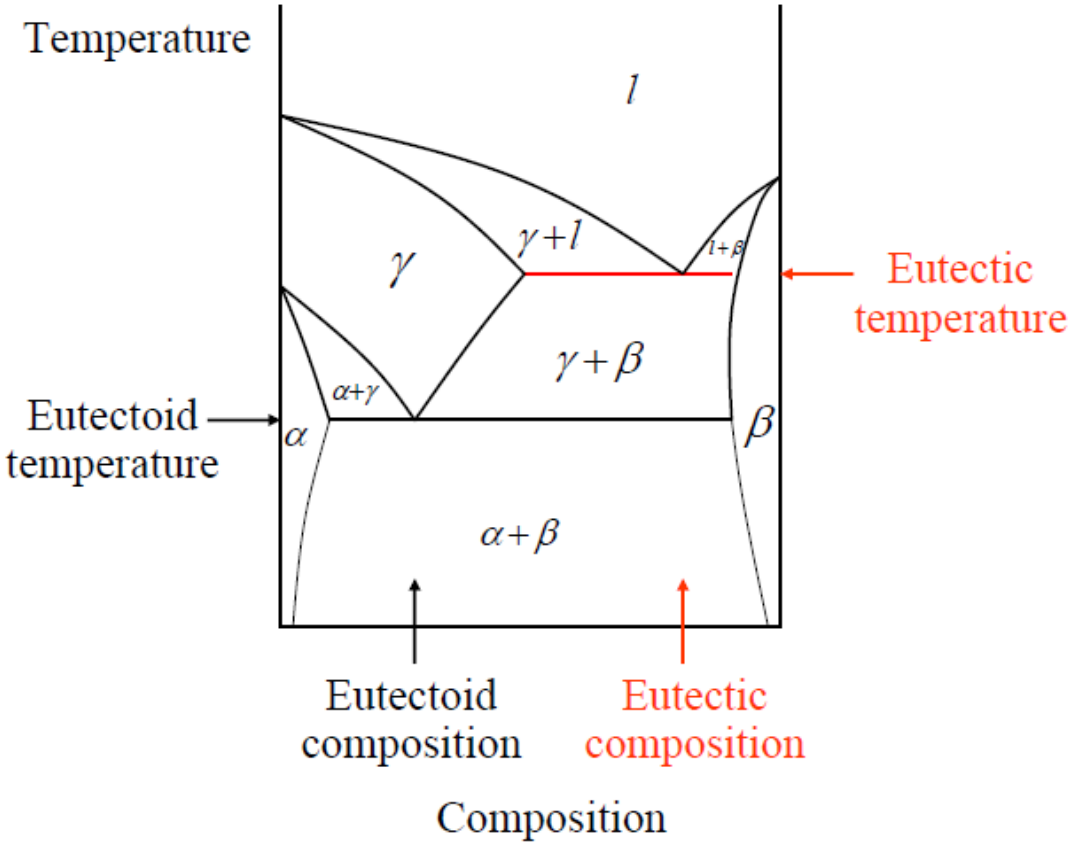
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Eutectoid Reactions



Eutectic and Eutectoid Reactions

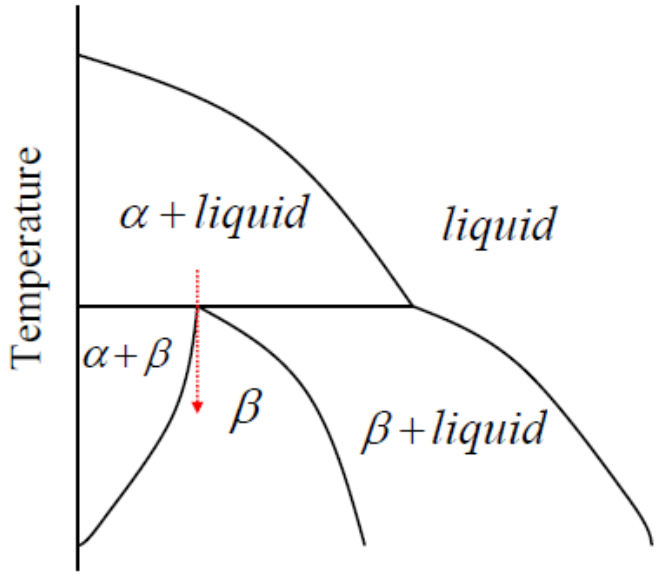
- The phase diagram below contains both an eutectic reaction and its solid-state analog, an eutectoid reaction.



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Peritectic Reactions

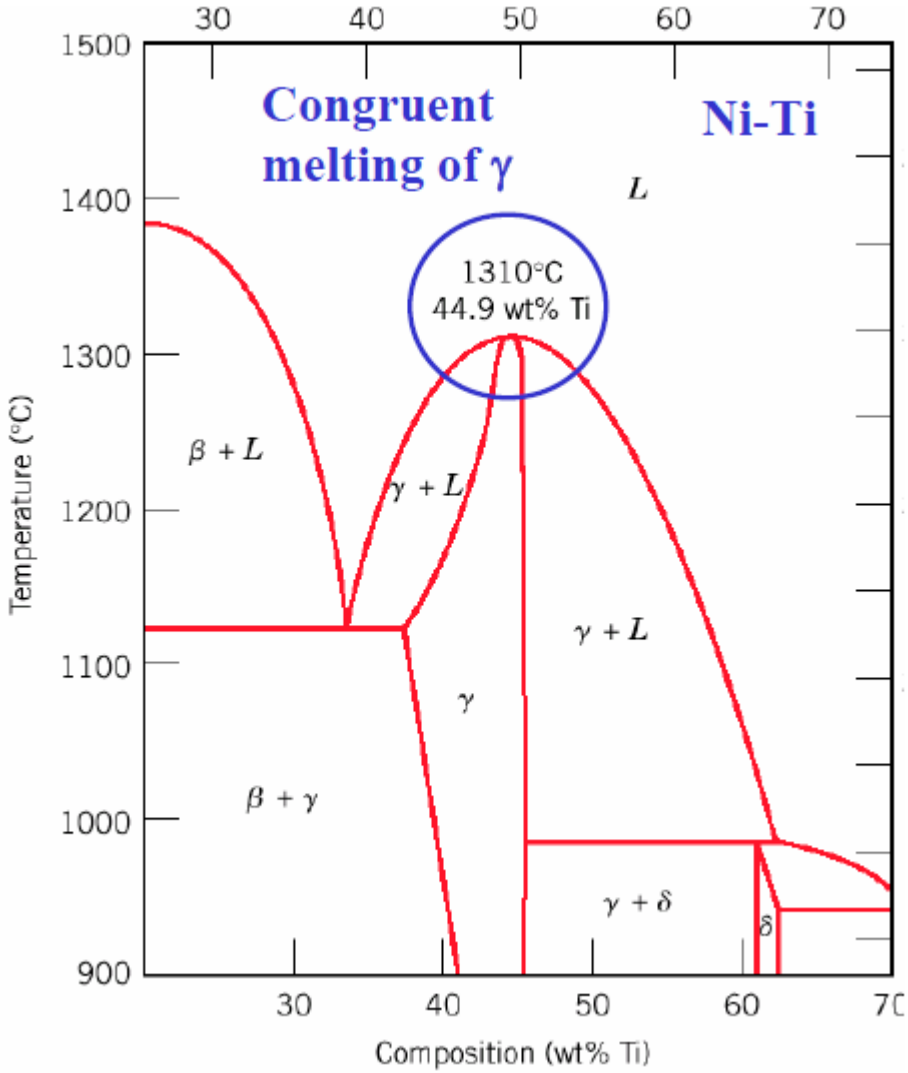
- A **peritectic** reaction - solid phase and liquid phase will together form a second solid phase at a particular temperature and composition upon cooling, e.g. L + α ↔ β These reactions are rather slow as the product phase will form at the boundary between the two reacting phases thus separating them, and slowing down any further reaction.
- Peritectics are not as common as eutectics and eutectiods, but do occur in some alloy systems.



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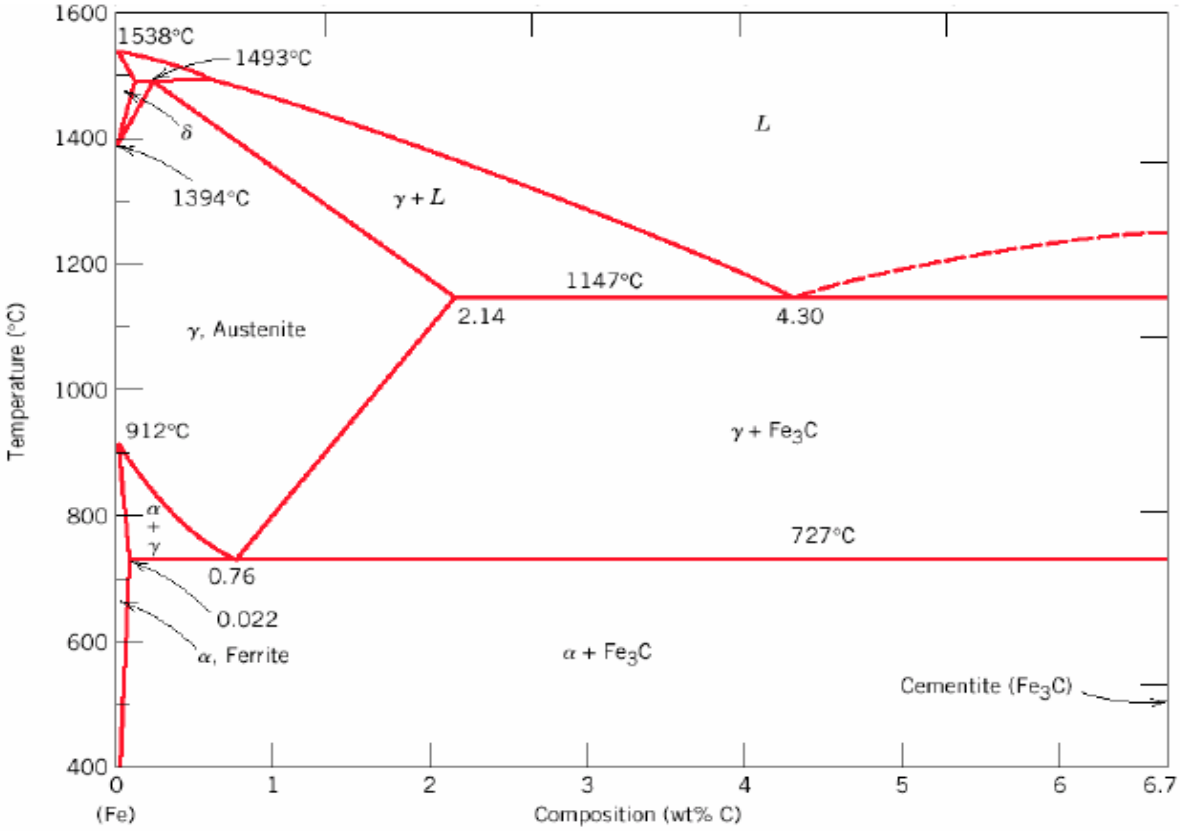
Congruent phase transformations

- **A congruent transformation** involves no change in composition (e.g., allotropic transformation such as α-Fe to γ-Fe or melting transitions in pure solids).
- For an incongruent transformation, at least one phase changes composition (e.g. eutectic, eutectoid, peritectic reactions).



The Iron–Iron Carbide (Fe–Fe_3C) Phase Diagram

- In their simplest form, steels are alloys of Iron (Fe) and Carbon (C). The Fe-C phase diagram is a fairly complex one, but we will only consider the steel part of the diagram, up to around 7% Carbon.



Phases in Fe–Fe_3C Phase Diagram

- **α-ferrite - solid solution of C in BCC Fe**
  - ✓ Stable form of iron at room temperature.
  - ✓ The maximum solubility of C is 0.022 wt%
  - ✓ Transforms to FCC γ-austenite at 912 °C
- **γ-austenite - solid solution of C in FCC Fe**
  - ✓ The maximum solubility of C is 2.14 wt %.
  - ✓ Transforms to BCC δ-ferrite at 1395 °C
  - ✓ Is not stable below the eutectoid temperature (727 ° C) unless cooled rapidly
- **δ-ferrite solid solution of C in BCC Fe**
  - ✓ The same structure as α-ferrite
  - ✓ Stable only at high T, above 1394 °C
  - ✓ Melts at 1538 °C

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The Fe–Fe_3C system

- C is an interstitial impurity in Fe. It forms a solid solution with α, γ, δ phases of iron
- Maximum solubility in BCC α-ferrite is limited (max. 0.022 wt% at 727 °C) - BCC has relatively small interstitial positions
- Maximum solubility in FCC austenite is 2.14 wt% at 1147°C - FCC has larger interstitial positions
- **Mechanical properties:** Cementite is very hard and brittle - can strengthen steels. Mechanical properties also depend on the microstructure, that is, how ferrite and cementite are mixed.
- **Magnetic properties:** α -ferrite is magnetic below 768 °C, austenite is non-magnetic.

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Classification of ferrous alloys

- **Iron:** less than 0.008 wt % C in α−ferrite at room T

- **Steels:** 0.008 - 2.14 wt % C (usually < 1 wt %)

  α-ferrite + Fe3C at room T

Examples of tool steel (tools for cutting other metals):

Fe + 1wt % C + 2 wt% Cr

Fe + 1 wt% C + 5 wt% W + 6 wt % Mo

Stainless steel (food processing equipment, knives, petrochemical equipment, etc.): 12-20 wt% Cr, ~$1500/ton

- **Cast iron:** 2.14 - 6.7 wt % (usually < 4.5 wt %)

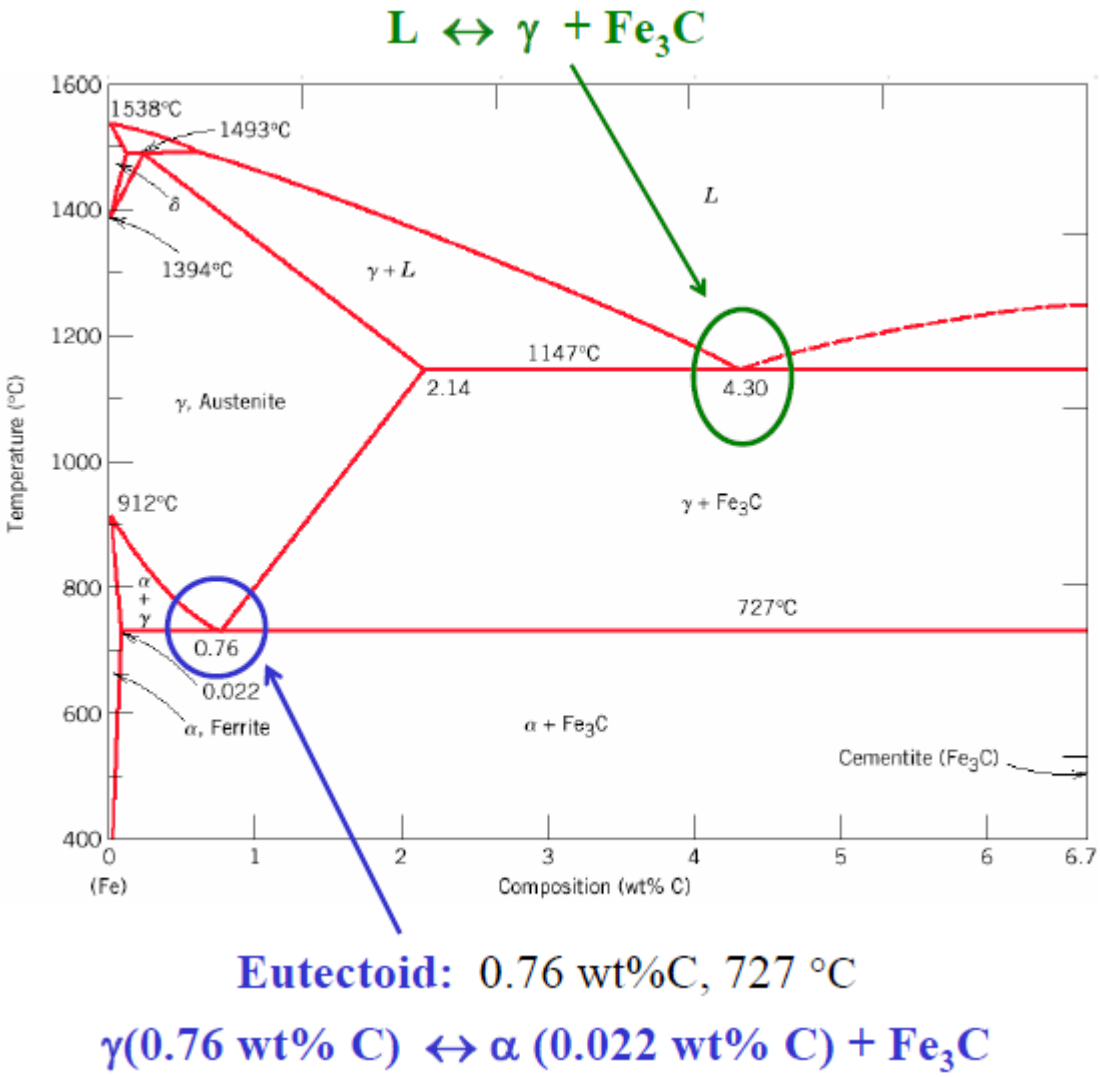
  heavy equipment casing

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Eutectic and eutectoid reactions in Fe–Fe_3C

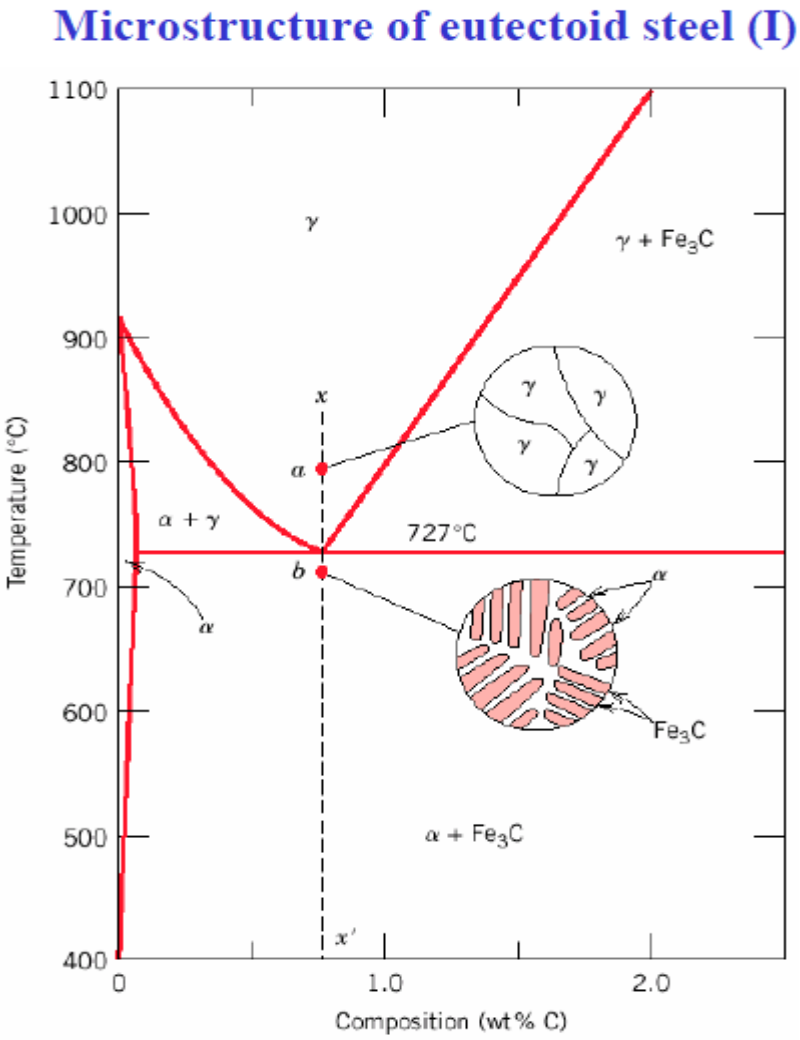
Eutectic: 4.30 wt% C, 1147 °C

- Eutectic and eutectoid reactions are very important in heat treatment of steels.



Development of Microstructure in Iron - Carbon alloys

- **Microstructure depends on composition (carbon content) and heat treatment.**
- In the discussion below we consider slow cooling in which equilibrium is maintained.



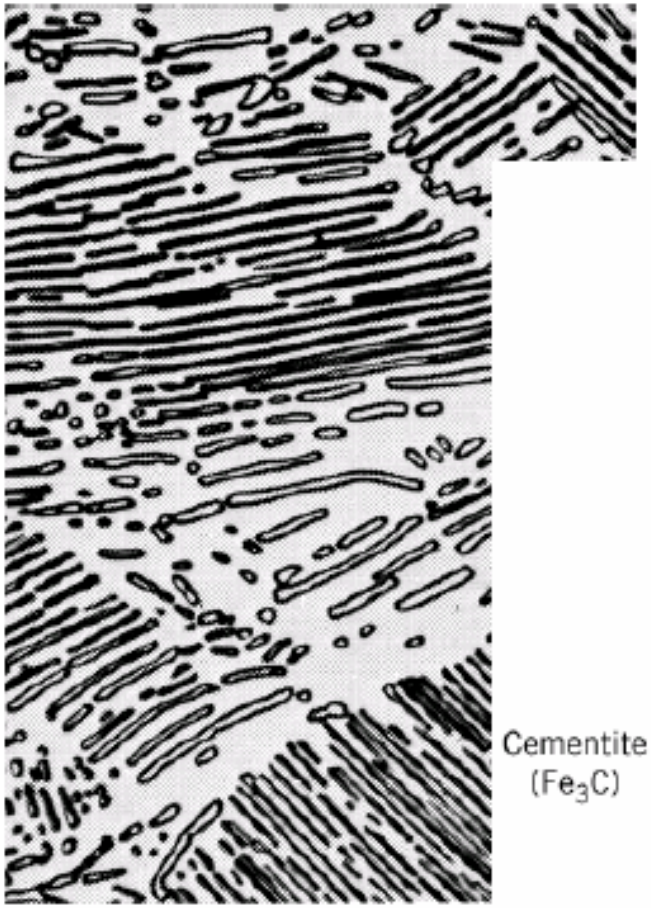
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Microstructure of eutectoid steel

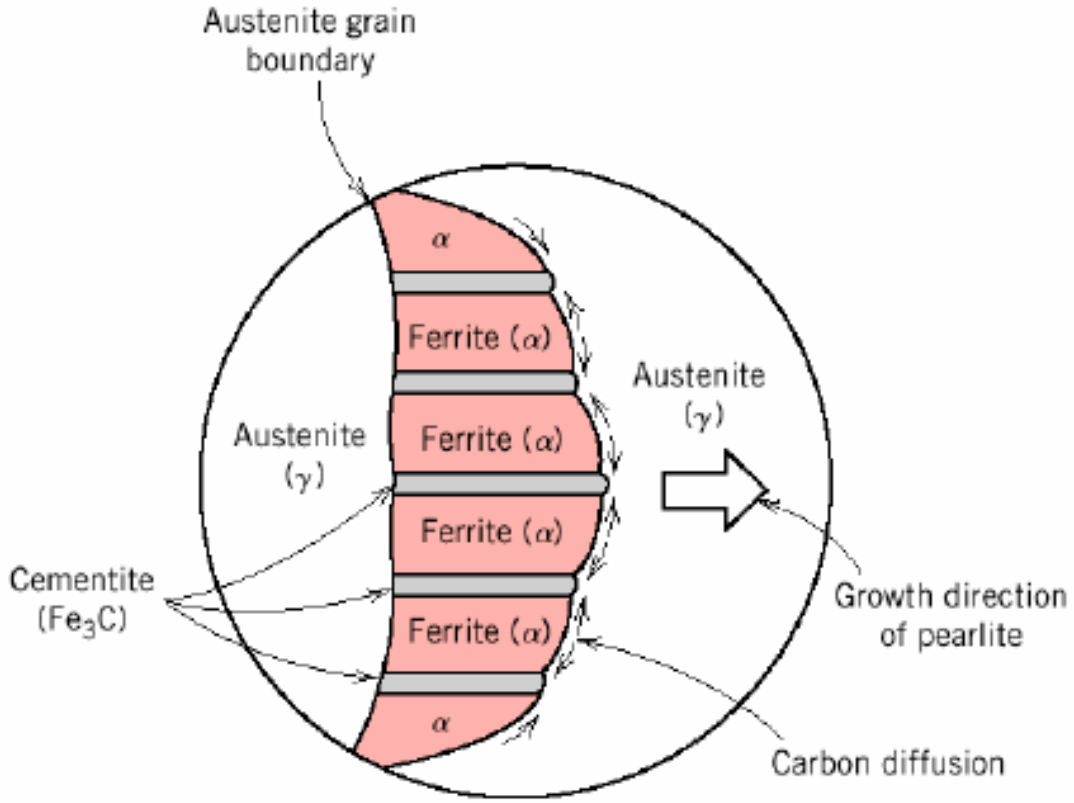
- When alloy of eutectoid composition (0.76 wt % C) is cooled slowly it forms **perlite**, a lamellar or layered structure of two phases: α-ferrite and cementite (Fe_3C)
- The layers of alternating phases in pearlite are formed for the same reason as layered structure of eutectic structures: redistribution C atoms between ferrite (0.022 wt%) and cementite (6.7 wt%) by atomic diffusion.
- Mechanically, pearlite has properties intermediate to soft, ductile ferrite and hard, brittle cementite.

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Microstructure of eutectoid steel

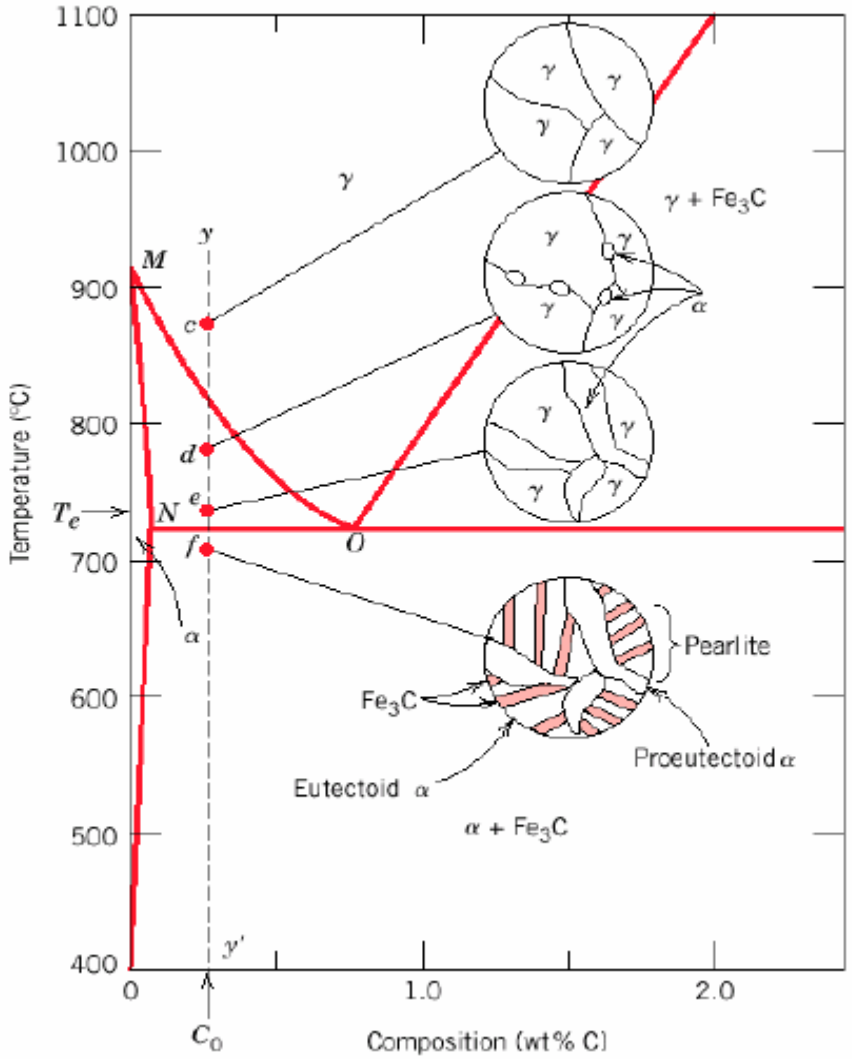


In the micrograph, dark areas are Fe_3C layers, light phase is α-ferrite



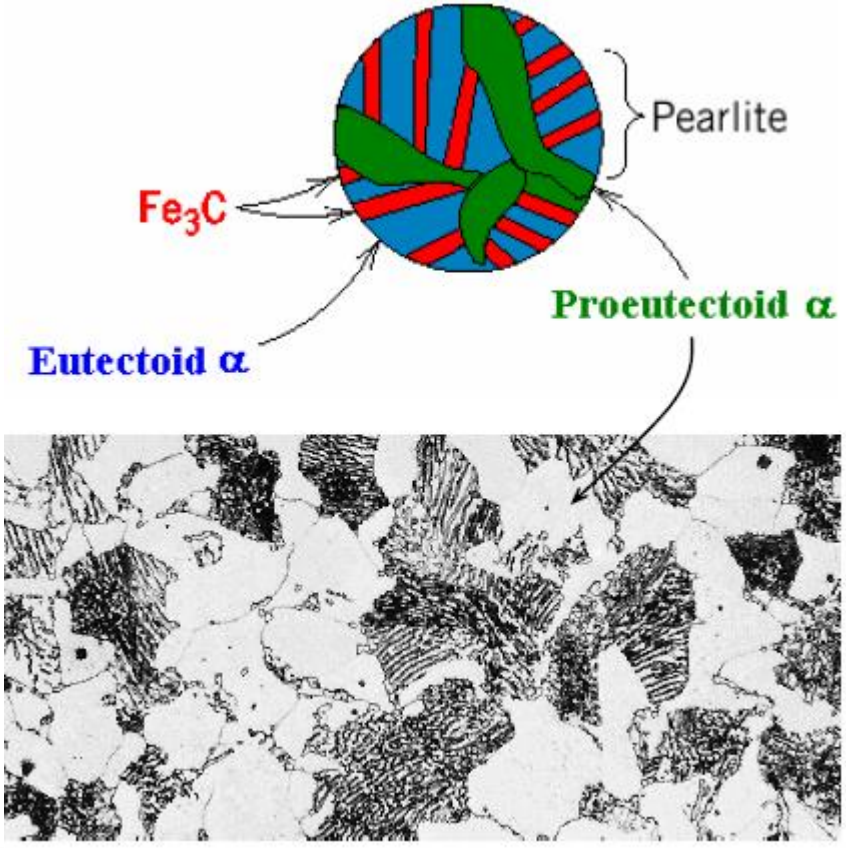
Microstructure of hypoeutectoid steel

- Compositions to the left of eutectoid (0.022 - 0.76 wt % C) **hypoeutectoid** (*less than eutectoid* -Greek) alloys.
- $\gamma \rightarrow \alpha + \gamma \rightarrow \alpha + Fe_3C$



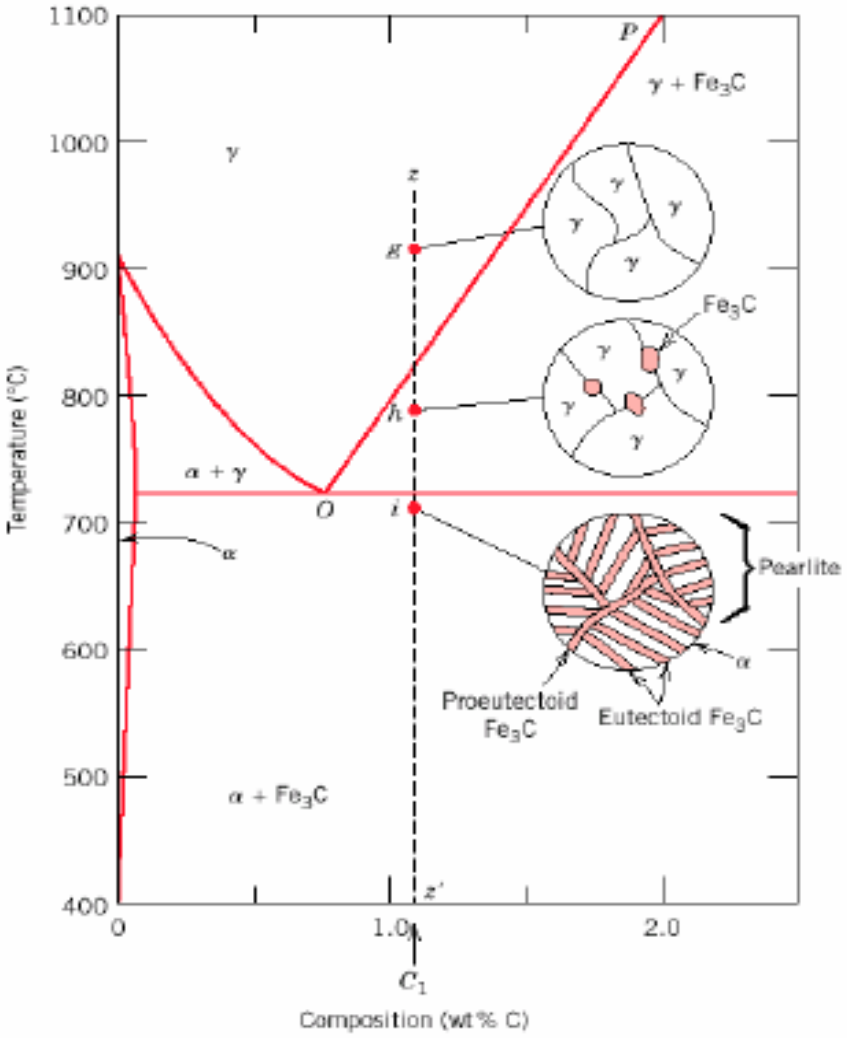
Microstructure of hypoeutectoid steel

- Hypoeutectoid alloys contain proeutectoid ferrite (formed above the eutectoid temperature) plus the eutectoid perlite that contain eutectoid ferrite and cementite.



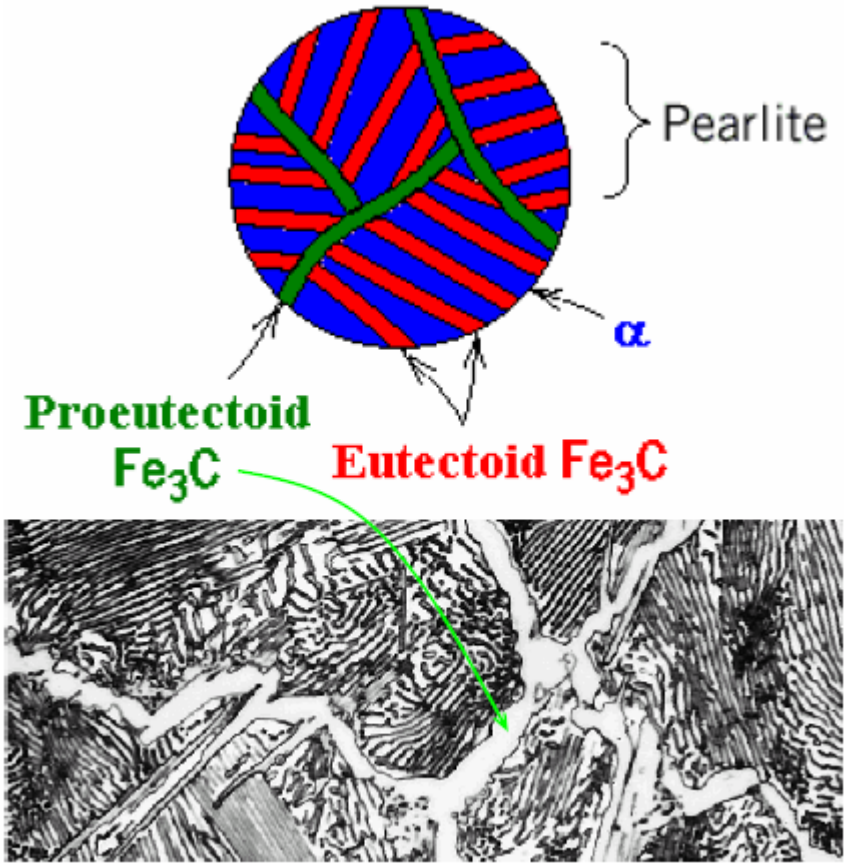
Microstructure of hypereutectoid steel

- Compositions to the right of eutectoid (0.76 - 2.14 wt % C) **hypereutectoid** (*more than eutectoid* -Greek) alloys.
- $\gamma \rightarrow \gamma + Fe_3C \rightarrow \alpha + Fe_3C$



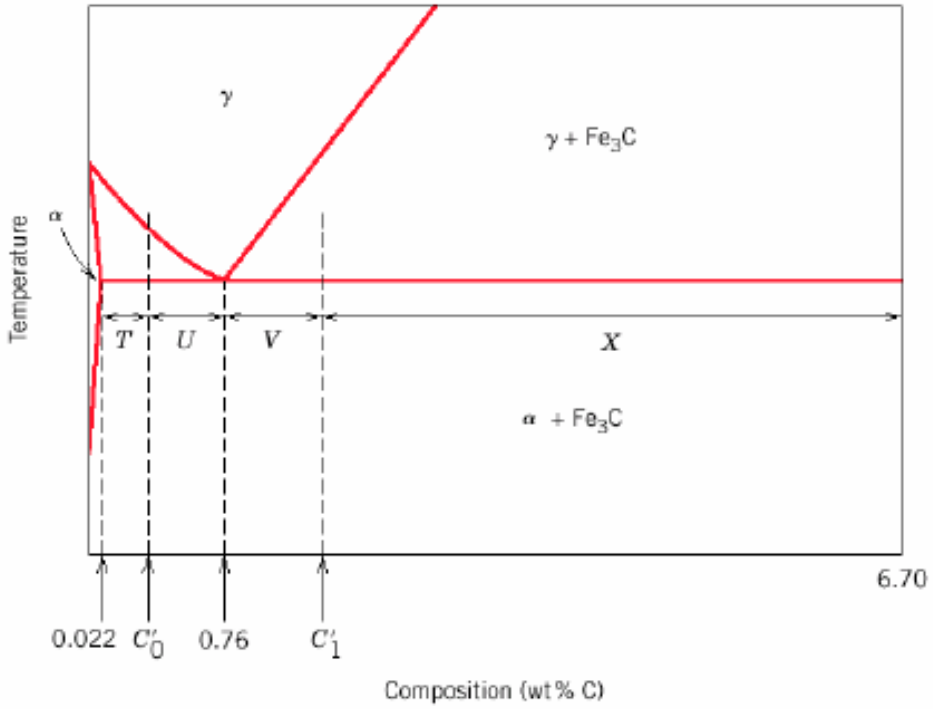
Microstructure of hypereutectoid steel

Hypereutectoid alloys contain proeutectoid cementite (formed above the eutectoid temperature) plus perlite that contain eutectoid ferrite and cementite.



How to calculate the relative amounts of proeutectoid phase (α or Fe_3C) and pearlite?

- Application of the lever rule with tie line that extends from the eutectoid composition (0.75 wt% C) to α – (α + Fe_3C) boundary (0.022 wt% C) for hypoeutectoid alloys and to (α + Fe_3C) – Fe_3C boundary (6.7 wt% C) for hypereutectoid alloys.
- Fraction of α phase is determined by application of the lever rule across the entire (α + Fe_3C) phase field.



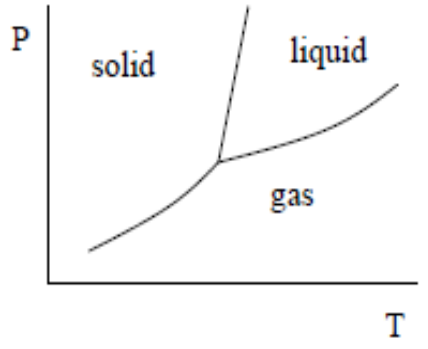
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The Gibbs phase rule

- Let's consider a simple one-component system.
- In the areas where only one phase is stable both pressure and temperature can be *independently* varied without upsetting the phase equilibrium → there are 2 **degrees of freedom**.
- Along the lines where two phases coexist in equilibrium, only one variable can be *independently* varied without upsetting the two-phase equilibrium (P and T are related by the Clapeyron equation) → there is only one degree of freedom.
- At the triple point, where solid liquid and vapour coexist any change in P or T would upset the three-phase equilibrium → there are no degrees of freedom.
- In general, the number of degrees of freedom, F, in a system that contains C components and can have P phases is given by the **Gibbs phase rule**:

$$P + F = C + N$$

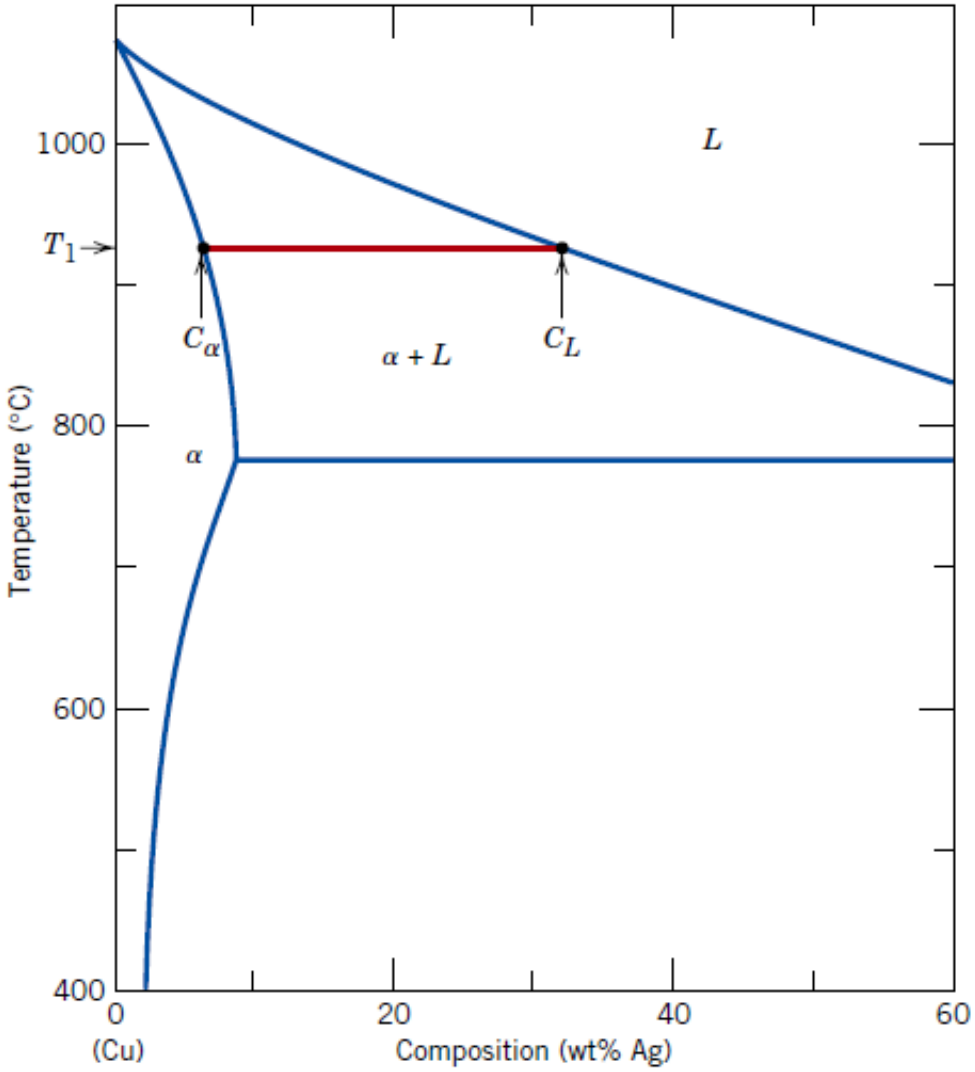
N.B: N is the number of non-compositional variables (e.g., temperature and pressure)



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The Gibbs phase rule – example of a binary system

- In one-phase regions of the phase diagram T and C can be changed independently.
- **In two-phase regions, F = 1. If the temperature is chosen independently, the compositions of both phases are fixed.**
- Three phases (L, α, β) are in equilibrium only at the eutectic point in this two component system (F = 0).



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Thank You